

COMPREHENSIVE REPORT

Credit Card Customer Segmentation

Unsupervised Clustering Analysis for Strategic Marketing & Service Delivery

Prepared By KARTIK RANA

Methodology Framework K-Means & Hierarchical Clustering, driven by
Exploratory Data Analysis and Feature Engineering

Date of Evaluation August 2026

Table of Contents

1. Executive Context	 3
2. The Business Challenge	 3
3. Strategic Objectives	 4
4. Data Description	 4
5. Overview of the Dataset	 5
Key Observations: SI_No and Customer Key	 5
Summary of the Dataset	 5
6. Exploratory Data Analysis	 6
Univariate Analysis	 6
Numerical Variables	6
<i>Avg_Credit_Limit</i>	6
<i>Total_Credit_Cards</i>	7
<i>Total_visits_bank</i>	8
<i>Total_visits_online</i>	9
<i>Total_calls_made</i>	10
CDF (Cumulative Distribution Function) Plots	 11
Labeled Barplots	 15
Feature Engineering Strategy (Avg_Credit_Limit)	 19
Bivariate Analysis	 22
The Density Jointplot: Channel Substitution	 26
Outlier Dashboard	 27
The "Cluster Hunter" Pairplot	 29
Synthesis of Key Patterns and Business Insights	 33
Key Meaningful Observations	 35
7. Data Preprocessing	 36
Missing Value Treatment	 36
Outlier Detection and Treatment	 36
Feature Selection: Removing Non-Predictive Variables	 38
Creating Derived Features (Channel Engagement Ratios)	 39
Skewness Mitigation	 40
Data Scaling (Standardisation)	 42

Table of Contents (continued)

8. Model Building: K-Means Clustering	 43
Methodology: Finding the Optimal K	43
Plotting the Elbow Curve	44
Checking Silhouette Scores	45
Silhouette Visualizer	46
Variance Decomposition: TSS, WSS and BSS	47
Final Model Execution	48
Cluster Profiling: Persona Identification	49
Visualising the Cluster Profiles	50
9. Model Building: Hierarchical Clustering	 51
Cophenetic Correlation Analysis	51
Dendrograms by Linkage Method	52
Determining the Optimal Number of Clusters	53
Hierarchical Cluster Profiling	54
10. K-Means versus Hierarchical Clustering	 55
Persona Stability Across Algorithms	55
Computation Time and Scalability	56
11. Actionable Business Recommendations	 57
For the Head of Marketing	57
For the Head of Delivery	57

List of Figures

Figure 1	Distribution and Boxplot of Average Credit Limit	6
Figure 2	Distribution and Boxplot of Total Credit Cards	7
Figure 3	Distribution and Boxplot of Total Bank Visits	8
Figure 4	Distribution and Boxplot of Total Online Visits	9
Figure 5	Distribution and Boxplot of Total Calls Made	10
Figure 6	Cumulative Distribution of Avg_Credit_Limit	11
Figure 7	Cumulative Distribution of Total_Credit_Cards	12
Figure 8	Cumulative Distribution of Total_visits_bank	12
Figure 9	Cumulative Distribution of Total_visits_online	12
Figure 10	Cumulative Distribution of Total_calls_made	13
Figure 11	Labeled Distribution of Total_Credit_Cards	16
Figure 12	Labeled Distribution of Total_visits_bank	16
Figure 13	Labeled Distribution of Total_visits_online	16
Figure 14	Labeled Distribution of Total_calls_made	17
Figure 15	Customer Wealth Distribution by Credit Limit Tier	20
Figure 16	Correlation Heatmap of Numerical Features	22
Figure 17	Channel Engagement by Customer Wealth Tier	24
Figure 18	Digital vs. Physical Channel Substitution	26
Figure 19	Outlier Identification Before Data Pre-processing	27
Figure 20	Detailed Bivariate Relationships (1 of 4)	29
Figure 21	Detailed Bivariate Relationships (2 of 4)	30
Figure 22	Detailed Bivariate Relationships (3 of 4)	31
Figure 23	Detailed Bivariate Relationships (4 of 4)	32
Figure 24	Channel Polarisation & Customer Wealth (4D View)	33
Figure 25	Distributions After 95th Percentile Capping	37
Figure 26	Distributions After Log Transformation	41
Figure 27	Elbow Method — Within-Cluster Sum of Squares by K	44
Figure 28	Silhouette Analysis — Average Score by K	45
Figure 29	Internal Cluster Quality — K = 3 compared with K = 4	46
Figure 30	Cluster Comparison — Financial and Channel Engagement Profiles	50
Figure 31	Hierarchical Clustering — Dendrograms by Linkage Method	52
Figure 32	Final Dendrogram (Average Linkage) with Cut Line at Distance = 3.0	53

List of Tables

Table 1	Data Dictionary	4
Table 2	Summary of the Dataset	5
Table 3	Missing Values Audit (Pre-processing Recheck)	36
Table 4	Preview of Engineered Interaction Features	39
Table 5	Scaled Data Overview (mean ≈ 0 , standard deviation ≈ 1)	42
Table 6	Variance Decomposition and WSS / BSS Ratio by Number of Clusters	47
Table 7	Cluster Assignment Preview	48
Table 8	Customer Profiles — Averages by Cluster	49
Table 9	Cophenetic Correlation by Distance Metric and Linkage Method	51
Table 10	Hierarchical Clustering — Segment Profiles (Averages)	54
Table 11	Persona Comparison — K-Means versus Hierarchical Clustering	55

Clustering Algorithms for Customer Segmentation

Complete Overview & Exploratory Data Analysis

1. Executive Context

Financial institutions face a continuous challenge in identifying and attracting the right target audience to remain competitive in the market. Recognizing that customer engagement and product optimization are critical to organizational success, banks frequently look for robust methodologies to deeply understand their customer base and improve market penetration.

AllLife Bank has observed that market penetration could be significantly enhanced. To achieve this, the Marketing team requires a structured approach to identify existing patterns within the customer base, ensuring that personalized campaigns accurately reflect the diverse financial profiles and behaviors of the clientele.

2. The Business Challenge

The core challenge for AllLife Bank is twofold. Firstly, the Marketing team proposes to run personalized campaigns to target new customers as well as upsell to existing customers, requiring granular customer segmentation. Secondly, market research has highlighted a critical operational issue: customers perceive the support services of the bank poorly. Based on this insight, the Operations team wants to upgrade the service delivery model to ensure that customer queries are resolved faster and through the appropriate channels. The Head of Marketing and Head of Delivery have therefore engaged the Data Science team to construct a data-driven solution.

3. Strategic Objectives

To address the demand for better market penetration and enhanced service delivery, AllLife Bank has implemented a data-driven, Machine Learning-based clustering solution. The primary goal is to develop a segmentation model capable of dividing the existing customers into distinct profiles based on their spending patterns and interaction history.

As a Data Scientist, the core deliverables of this analysis include:

- **Streamline Operations:** Facilitate the service delivery model upgrade by identifying how different customer segments prefer to interact with the bank (in-person, online, or telephonic), thereby reducing query resolution bottlenecks and improving support perception.
- **Identify Key Drivers:** Analyze historical data to determine the variables and financial behaviors that most significantly define a customer's specific profile and channel preference.
- **Profile Recommendation:** Formulate data-driven recommendations outlining suitable targeted marketing campaigns to optimize upselling to existing customers and accurately target new prospects.

4. Data Description

The data contains various attributes regarding the customers' financial status and their interaction methods with the bank. The detailed data dictionary is provided below:

Attribute	Description
SI_No	Primary key of the records
Customer Key	Customer identification number
Avg_Credit_Limit	Average credit limit of each customer for all credit cards
Total_Credit_Cards	Total number of credit cards possessed by the customer
Total_visits_bank	Total number of visits that the customer made personally to the bank
Total_visits_online	Total number of visits or online logins made by the customer
Total_calls_made	Total number of calls made by the customer to the bank or its customer service department

Table 1: Data Dictionary

5. Overview of the Dataset

✦ **Data Volume:** There are 660 observations and 7 columns in the dataset.

✦ **Data Type:** All 7 features are of numerical type.

Key Observations: SI_No and Customer Key

- **Nature of Features:** Both variables are strictly administrative identifiers and completely lack behavioral, financial, or demographic predictive value.
- **Risk to Models:** Retaining them introduces mathematical noise, as distance-based algorithms (like K-Means) would incorrectly interpret these arbitrary, sequential IDs as meaningful numerical weights, distorting the final clusters.
- **Action:** Because they offer no analytical value, both features will be permanently dropped during the initial data cleaning phase.

Summary of the Dataset

	count	mean	std	min	25%	50%	75%	max
SI_No	660.0	330.500000	190.669872	1.0	165.75	330.5	495.25	660.0
Customer Key	660.0	55141.443939	25627.772200	11265.0	33825.25	53874.5	77202.50	99843.0
Avg_Credit_Limit	660.0	34574.242424	37625.487804	3000.0	10000.00	18000.0	48000.00	200000.0
Total_Credit_Cards	660.0	4.706061	2.167835	1.0	3.00	5.0	6.00	10.0
Total_visits_bank	660.0	2.403030	1.631813	0.0	1.00	2.0	4.00	5.0
Total_visits_online	660.0	2.606061	2.935724	0.0	1.00	2.0	4.00	15.0
Total_calls_made	660.0	3.583333	2.865317	0.0	1.00	3.0	5.00	10.0

Table 2: Summary of the Dataset

6. Exploratory Data Analysis

Univariate Analysis

Numerical Variables

Avg_Credit_Limit

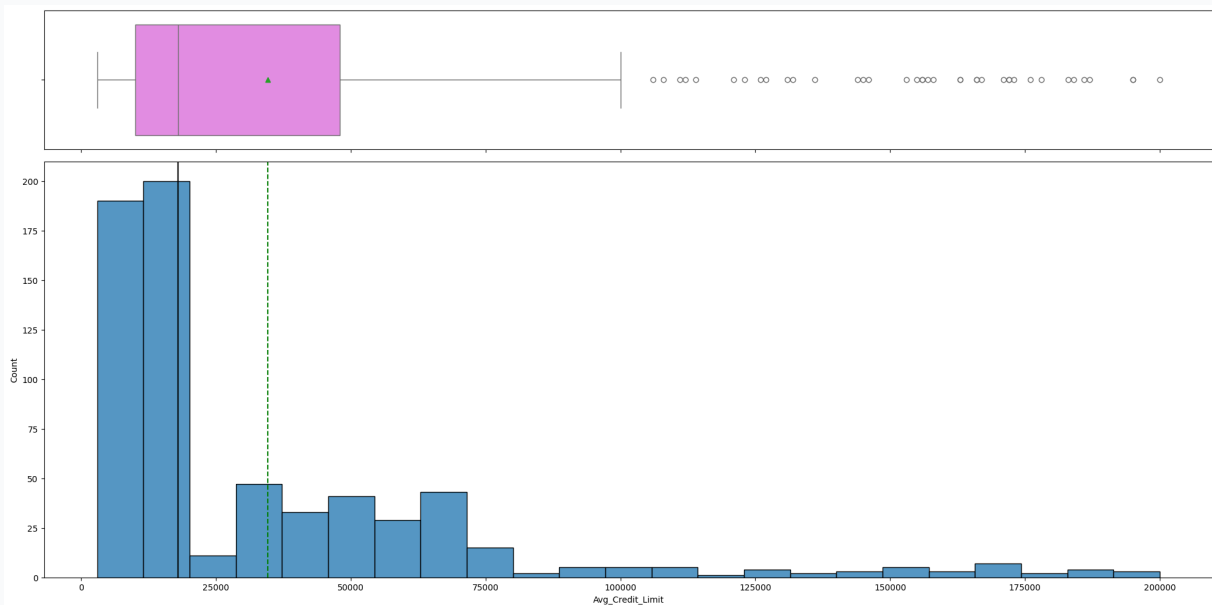


Figure 1: Distribution and Boxplot of Average Credit Limit

Key Observations: Avg_Credit_Limit

- **Sharp frequency drop-off:** the customer base is heavily concentrated (~400 individuals) in credit limits under 25,000, causing an abrupt frequency drop rather than a gradual taper.
- **Dense outlier population:** boxplot outliers starting around 100,000 form a continuous line, highlighting a solid, consistent sub-segment rather than rare data anomalies.
- **Strategic extreme values:** these upper-end values represent a genuine, affluent cohort, not data errors. They must be preserved for accurate segmentation and targeted marketing.
- **Tiered limit assignments:** clusters between 100,000 and 200,000 reflect standard banking practice of assigning premium limits in rounded tiers (e.g. 100k, 150k, 175k).
- **Clustering and pre-processing:** to stop extreme values disproportionately skewing distance-based algorithms like K-Means, a logarithmic transformation or robust scaling is essential before model training.

Total_Credit_Cards

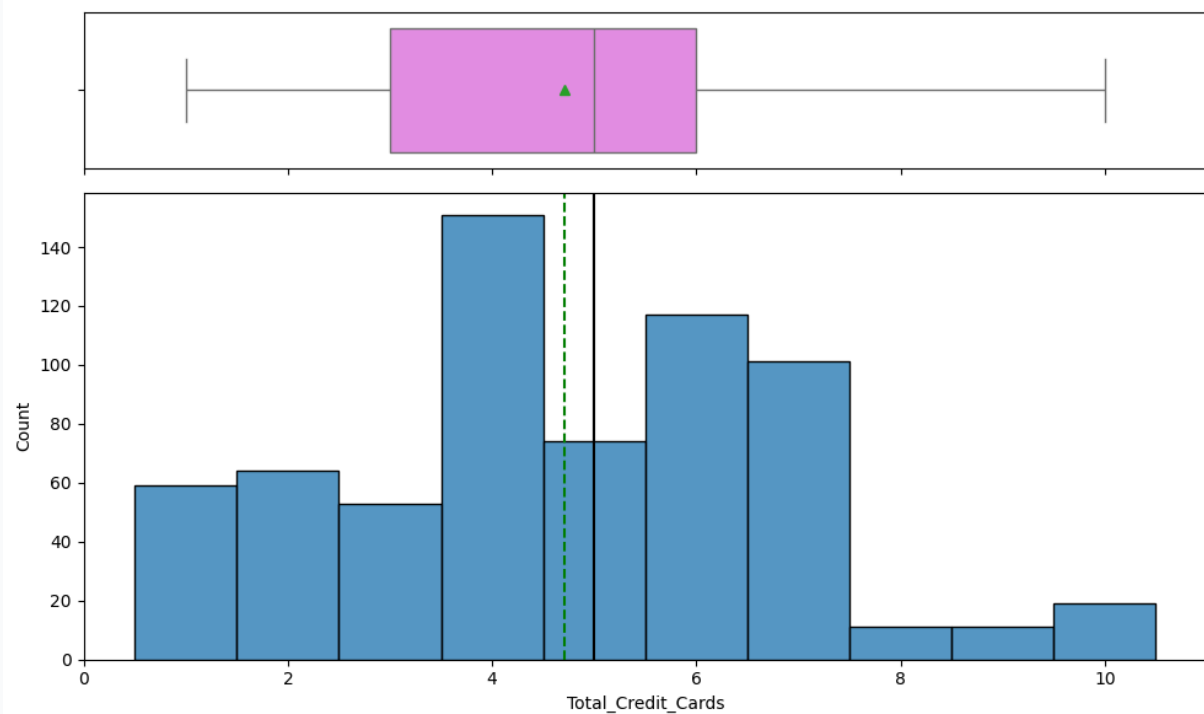


Figure 2: Distribution and Boxplot of Total Credit Cards

Key Observations: Total_Credit_Cards

- **Distributional characteristics:** a slight left skew is evident, with the mean (4.71) sitting marginally below the median (5.0). The customer base therefore leans toward holding five or more credit cards.
- **Data spread and variability:** a standard deviation of 2.17 reflects a tight, expected spread, with most customers falling within the interquartile range of 3 to 6 cards.
- **High-end values (business context):** the maximum holding is capped at 10 cards. In a business context, such volume is a strong indicator of high earners with significant spending power and excellent creditworthiness.
- **Outlier treatment not required:** a maximum of 10 credit cards is a valid, logically sound value in a real-world banking setting rather than an error, so outlier capping or removal is unnecessary for this column.
- **Pre-processing recommendation:** this feature operates on a constrained 1 to 10 scale, whereas Avg_Credit_Limit extends to 200,000. Because distance-based algorithms such as K-Means are highly sensitive to magnitude, standard or min-max scaling across all numerical features is strictly necessary to prevent high-magnitude variables from dominating the distance calculations and distorting the final clusters.

Total_visits_bank

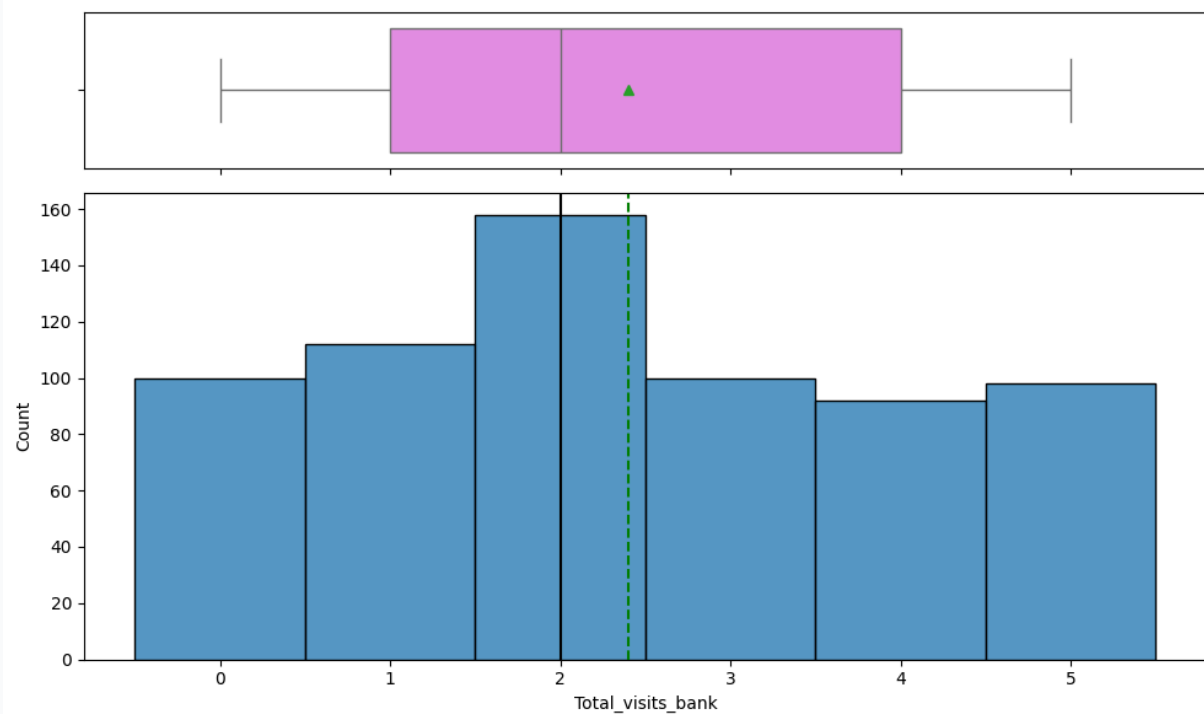


Figure 3: Distribution and Boxplot of Total Bank Visits

Key Observations: Total_visits_bank

- **Central tendency and peak:** the histogram reveals a prominent mode at two visits per year, the most common behaviour in the customer base. The mean of roughly 2.4 sits slightly to the right of the median of 2, indicating a very slight rightward skew.
- **Data spread and absence of outliers:** the boxplot shows the data cleanly bounded between 0 and 5 visits, with whiskers extending to the absolute minimum and maximum. There are no statistical outliers in this feature.
- **Interquartile range:** the middle 50% of the customer base visits a physical branch between one and four times annually.
- **The zero-visit segment:** the bar at 0 shows that a significant cohort of roughly 100 customers never visits a branch, visibly confirming a distinct segment that has fully substituted in-person banking with remote channels.
- **Low engagement ceiling:** branch visits cap at 5. This low ceiling emphasises that in-person visits are relatively infrequent across the entire portfolio, signalling a broader behavioural shift toward digital or telephonic query resolution.

Total_visits_online

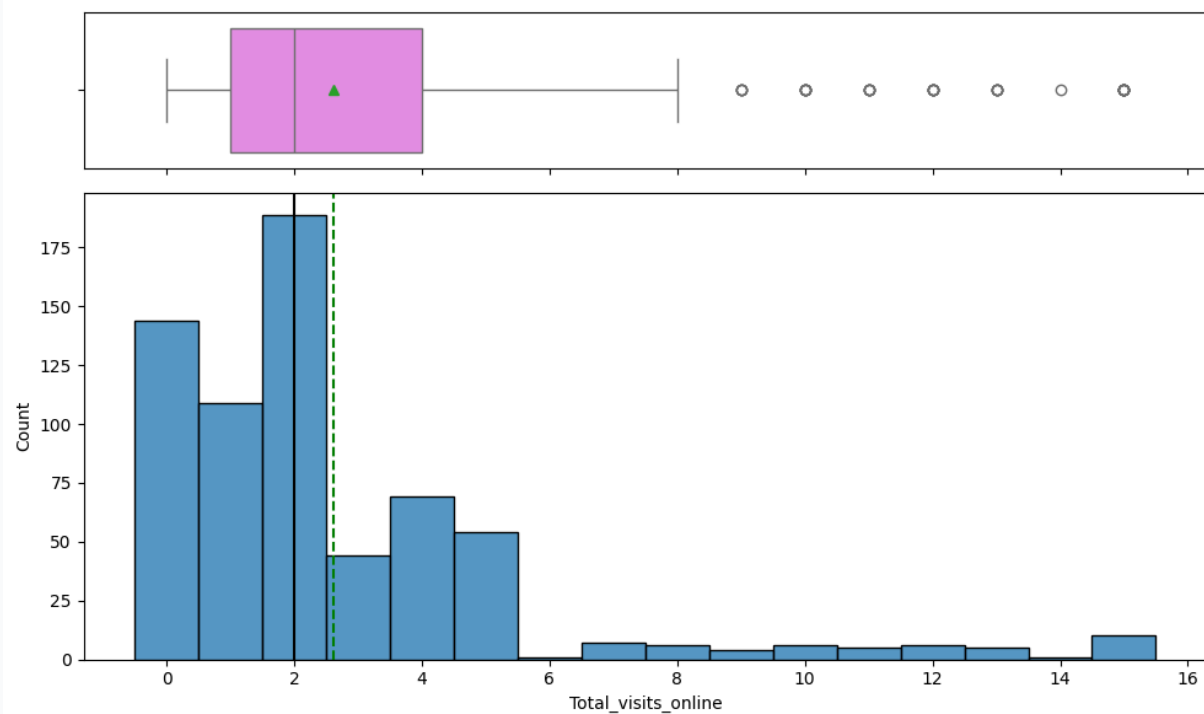


Figure 4: Distribution and Boxplot of Total Online Visits

Key Observations: Total_visits_online

- **Central tendency and right skew:** the distribution is distinctly right-skewed. The median is 2 online visits, but a long tail of high-frequency users pulls the mean higher, to approximately 2.6.
- **Primary peak (mode):** the most common behaviour is exactly two online visits per year, forming the highest peak in the histogram at 189 customers.
- **The zero-digital segment:** a substantial cohort of 144 customers records exactly zero online visits. This reinforces the channel substitution theory — a large part of the portfolio bypasses digital banking entirely, relying on phone calls or branch visits.
- **High-engagement long tail:** frequency drops steeply after five online visits, but a prominent tail extends all the way to 15. Unlike bank visits, which capped at 5, this channel has power users: the small but consistent bars from 7 to 15 visits mark a highly engaged, digital-first segment.
- **Clustering implications:** the wide variance in this feature — from zero to 15 — makes it an excellent differentiator, cleanly separating traditional offline customers from digital-heavy power users.

Total_calls_made

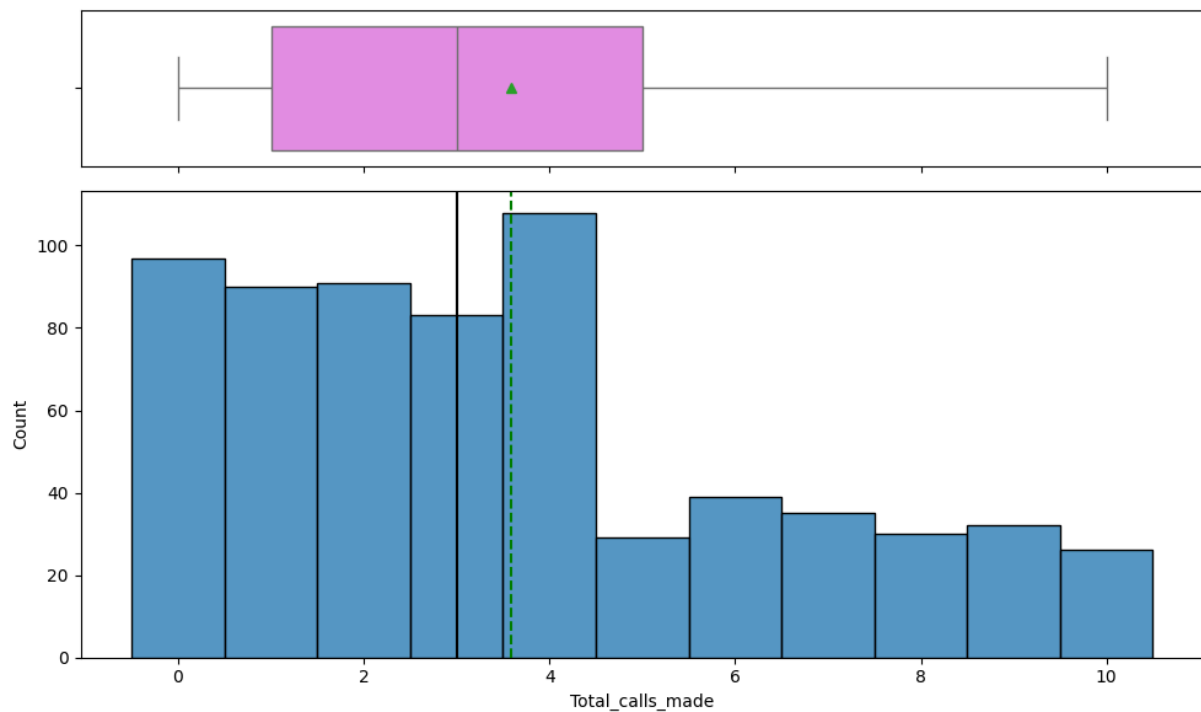


Figure 5: Distribution and Boxplot of Total Calls Made

Key Observations: Total_calls_made

- **Central tendency and skewness:** the distribution is right-skewed. The median is 3 calls, while the mean is pulled higher to roughly 3.6 by high-frequency callers in the tail.
- **Data spread (IQR):** the middle 50% of the customer base makes between one and five calls to the bank annually.
- **Primary peak (mode):** the highest frequency is exactly four calls per year, though volume is fairly evenly distributed across the 0 to 4 range before dropping off significantly.
- **The zero-call segment:** a substantial cohort of nearly 100 customers records exactly zero calls, again reinforcing channel substitution — this group bypasses telephone support entirely, relying on digital portals or branch visits.
- **High-engagement segment and absence of outliers:** after the drop at five calls there is a consistent, flat plateau of customers making between five and 10 calls. The boxplot reveals no statistical outliers beyond the upper whisker, so a maximum of 10 calls sits within the expected behavioural spread — a dedicated segment that relies heavily on direct verbal communication.

CDF (Cumulative Distribution Function) Plots

Understanding the CDF Plot

An Empirical Cumulative Distribution Function (ECDF) plot visualises the overall distribution of numerical data. Unlike histograms, which display frequency grouped into arbitrary bins, a CDF plot represents the cumulative proportion of the dataset up to a given value.

- **X-axis:** the actual continuous or discrete values of the numerical feature.
- **Y-axis:** the cumulative proportion of the customer base, from 0.0 (0%) to 1.0 (100%).

Key Advantages for Segmentation Analysis

- **Direct percentile identification:** exact percentiles (the median at 0.5, the 75th percentile at 0.75) can be read straight off the curve by tracing a horizontal line from the y-axis, bypassing the need for separate statistical tables.
- **Absence of binning bias:** CDF plots map every individual data point, giving the truest, unaltered representation of the data's underlying shape and spread.
- **Interpreting distribution shape:** a steep vertical climb indicates a high concentration of customers within a narrow range of values; a gradual, flat slope signifies wide dispersion across a broader range.

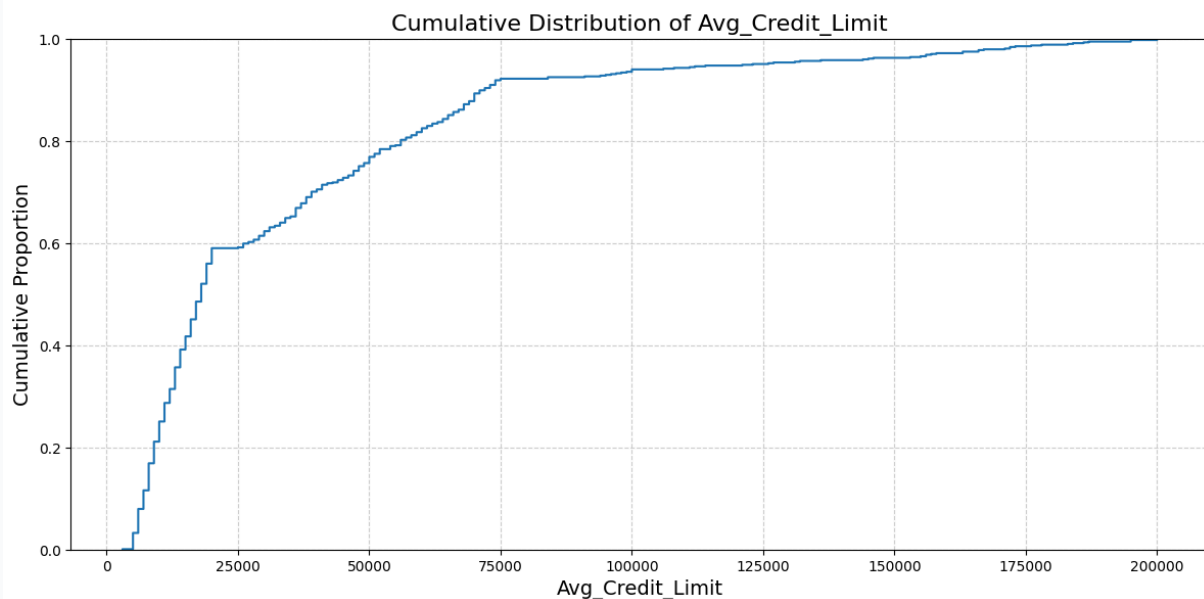


Figure 6: Cumulative Distribution of Avg_Credit_Limit

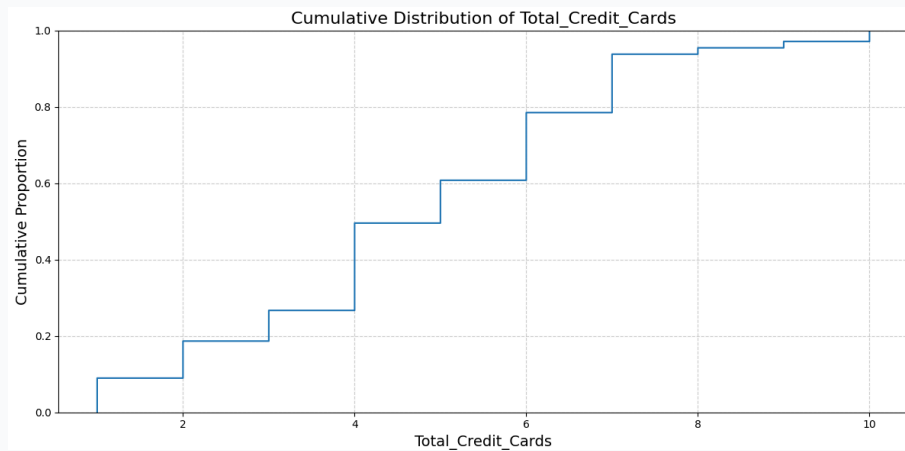


Figure 7: Cumulative Distribution of Total_Credit_Cards

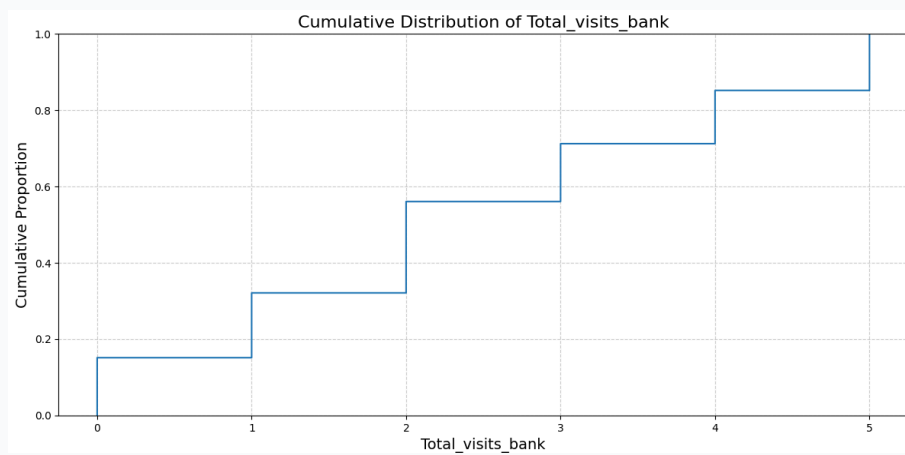


Figure 8: Cumulative Distribution of Total_visits_bank

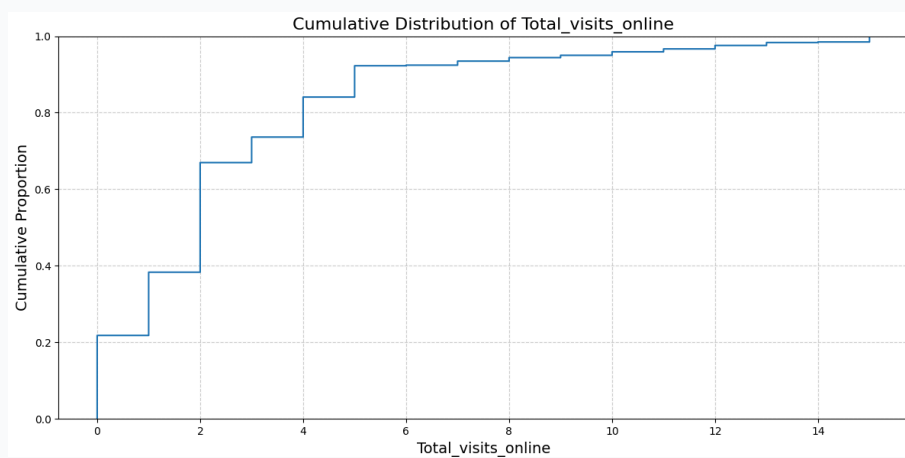


Figure 9: Cumulative Distribution of Total_visits_online

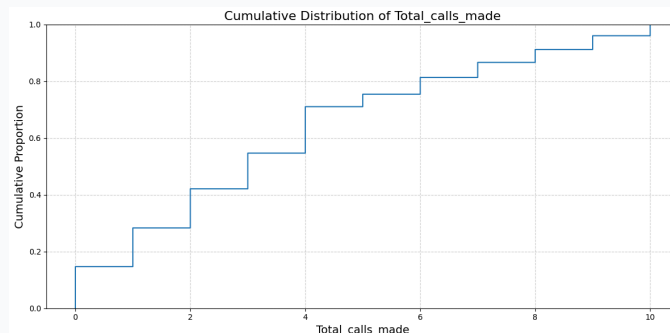


Figure 10: Cumulative Distribution of Total_calls_made

Important Patterns and Observations

The CDF plots give a clear picture of customer density and behavioural dispersion across the portfolio. Because four of the five features are discrete counts, their curves appear as staircases, where the height of each step is the exact proportion of the customer base exhibiting that specific behaviour.

1. Avg_Credit_Limit (the wealth distribution)

- **Rapid initial ascent:** the curve climbs extremely fast on the far left, confirming that roughly 75% of the customer base is densely concentrated in the lower credit tiers, below 50,000.
- **The affluent plateau:** past the 100,000 mark the curve flattens into a long horizontal plateau extending to 200,000 — a highly dispersed tail representing a small, exclusive segment holding exceptionally high limits.

2. Total_Credit_Cards (product holding)

- **Concentrated core:** the steepest vertical steps occur between three and six cards, placing the core of the portfolio in this mid-range. Very few customers hold just one card, and only a minority reach the maximum of 10.
- **Predictable spread:** the curve climbs steadily without extreme horizontal gaps, showing card ownership is well distributed and free of massive outliers.
- **Absolute peak at four cards:** the single steepest step occurs precisely at the four-card mark, signifying the mode of the distribution — the most heavily concentrated individual portfolio size.

3. Total_visits_bank (the physical branch channel)

- **The zero-in-person step:** the first vertical step at 0 reveals that about 15% of the portfolio never visits a branch.
- **The dominant mode:** the tallest single step occurs at two visits, confirming this as the most widespread branch engagement pattern.
- **Early cap:** the curve reaches 1.0 very quickly, at just five visits. The absence of a long tail proves high-frequency in-person banking is practically non-existent here.

4. Total_visits_online (the digital channel)

- **Massive zero-digital step:** as with bank visits there is a large initial step at 0, visualising a distinct cohort that entirely bypasses online banking portals.
- **The power-user tail:** unlike branch visits, this curve does not cap early. A long staircase of smaller steps stretches to 15 visits, proving the existence of a highly engaged, digital-first segment that interacts with the bank far more frequently than offline customers.

5. Total_calls_made (the telephone channel)

- **Evenly distributed core:** the steps between one and four calls are relatively uniform in height, showing a steady distribution of customers who rely on phone support for standard, infrequent enquiries.
- **The zero-call step:** a noticeable step at 0 confirms a segment that completely substitutes phone calls with other channels, such as the online portal.
- **Moderate tail:** the curve tapers smoothly toward a maximum of 10 calls. The smaller steps in the 5 to 10 range highlight a minority of high-maintenance customers who require significant verbal support.

Overall Conclusion for Clustering

The CDF plots vividly validate the phenomenon of **channel substitution**. The distinct zero-engagement steps at the start of the bank, online and call plots prove that customers actively choose preferred mediums while completely ignoring others.

This high variance in behavioural preference will serve as an excellent mathematical differentiator, allowing the clustering algorithm to identify actionable customer segments.

Labeled Barplots

Understanding Labeled Barplots

Labeled barplots display the frequency or proportion of discrete numerical data, augmented with exact numerical labels directly on each bar. In this analysis the bars represent the exact percentage of the customer base falling into each behavioural category.

Strategic Value of Labeled Barplots

- **Exact quantification:** unlike CDFs or standard histograms, labeled barplots directly quantify the exact percentage or count for each category. Rather than estimating from a curve, they state the precise proportion of customers holding exactly four credit cards, providing concrete, reportable metrics.
- **Optimal for discrete data:** the features analysed here (Total_Credit_Cards, Total_visits_bank, Total_visits_online, Total_calls_made) are discrete. Treating them as distinct categories within a bar chart is the cleanest visualisation, avoiding the implication of continuous fractions such as 2.5 bank visits that can occur with density plots.

Strategic Purpose for the Business Context

For this segmentation project, labeled barplots act as a bridge between data science and business strategy:

- **Stakeholder clarity:** density plots and cumulative distributions are excellent for pre-processing and algorithm preparation, but business leaders need immediate, concrete figures. These plots translate abstract statistical shapes into presentation-ready metrics.
- **Quantifying segment opportunities:** exact percentages let stakeholders size target cohorts accurately. Knowing the precise share of customers holding a given number of cards helps Marketing forecast the reach and budget of an upselling campaign.
- **Validating channel substitution:** the plots give hard percentages for the zero-engagement groups across each service channel, quantifying exactly how much of the portfolio has abandoned physical branches in favour of digital or telephonic support.

Labeled Barplots

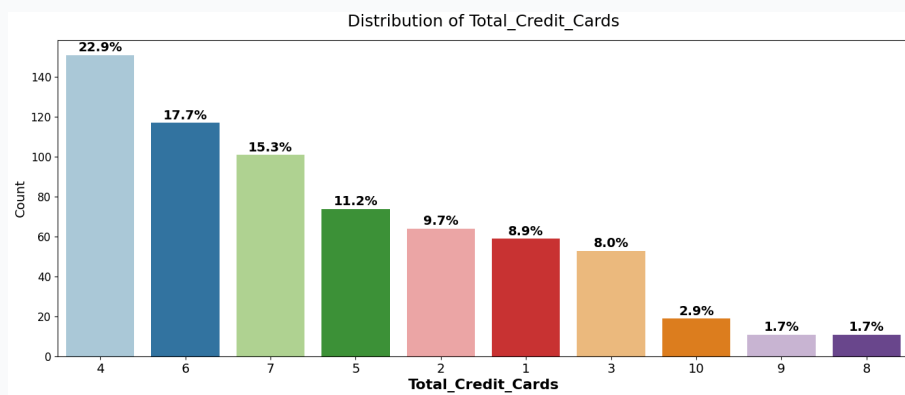


Figure 11: Labeled Distribution of Total_Credit_Cards

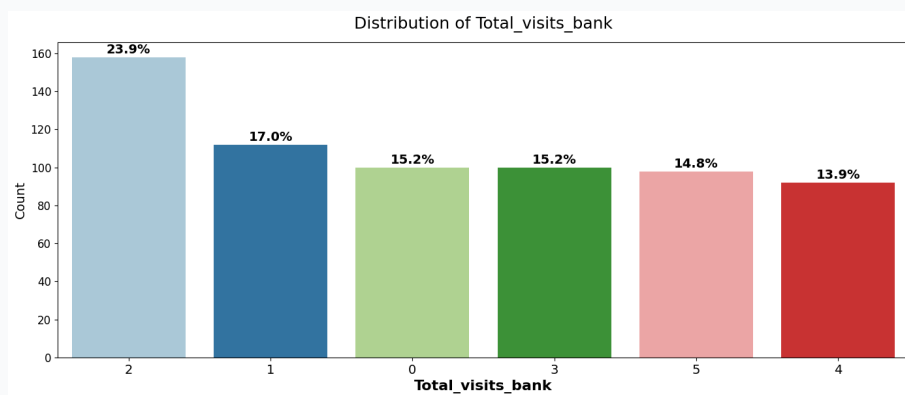


Figure 12: Labeled Distribution of Total_visits_bank

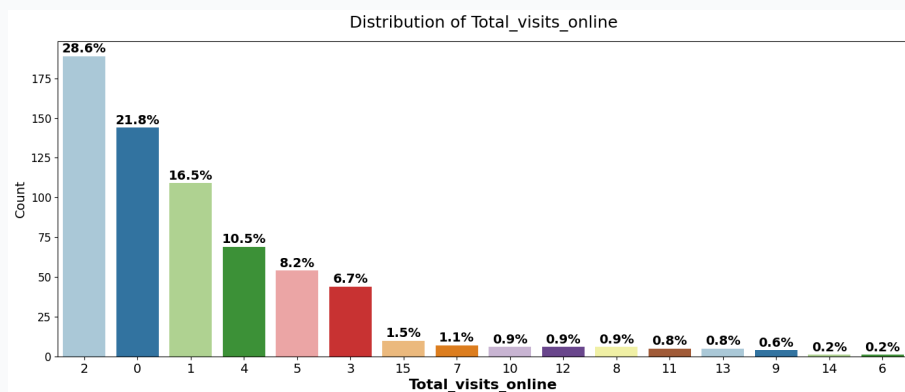


Figure 13: Labeled Distribution of Total_visits_online

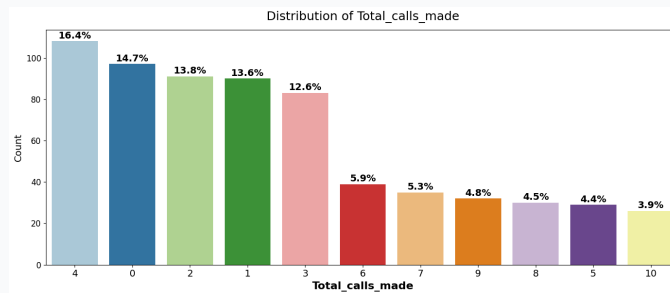


Figure 14: Labeled Distribution of Total_calls_made

Important Patterns & Business Observations

The combined labeled barplots give a quantified, presentation-ready view of customer behaviour. Evaluating the exact percentage distribution across product holdings and engagement channels identifies the baseline metrics required for targeted marketing and service optimisation.

1. Product Holding (Total_Credit_Cards)

- **The modal customer:** the single most common portfolio size is exactly four credit cards, at 22.9%.
- **Concentrated core:** the vast majority of the customer base holds between three and seven cards, forming a stable, predictable core.
- **Marketing implication:** this high concentration in the middle tiers gives the Head of Marketing a large, readily available audience for campaigns upselling four-card holders into premium five-plus tiers.

2. Physical Branch Engagement (Total_visits_bank)

- **Low dependency:** in-person banking shows the lowest engagement ceiling of any channel, hard-capping at five visits per year.
- **Zero-engagement cohort:** a significant 15.2% of the portfolio records no branch visits annually.
- **Service implication:** physical branch dependency is demonstrably low, validating a strategic shift toward upgrading remote service channels rather than expanding the physical footprint.

3. Digital Channel Dominance (Total_visits_online)

- **The power users:** unlike physical visits, the digital channel carries a highly populated long tail stretching to 15 visits per year, marking a highly engaged, digital-first segment.
- **The digital holdouts:** conversely, 21.8% of customers record exactly zero online visits, indicating complete digital abandonment.
- **Service implication:** this extreme variance signals that while digital dominates for one group, the bank cannot fully deprecate alternative support methods without alienating the zero-digital cohort.

4. Telephonic Support (Total_calls_made)

- **Balanced core volume:** engagement is fairly evenly distributed across the one to four call range, representing standard, periodic customer enquiries.
- **High-maintenance segment:** a consistent, measurable percentage of customers make up to 10 calls annually — a high-touch segment that relies heavily on verbal communication.
- **Service implication:** this is crucial for the Operations team in forecasting call-centre staffing and optimising query resolution times for this highly vocal segment.

Crucial Takeaway for Clustering: Channel Substitution

Across all three service channels — bank, online and call — a distinct zero-engagement bar appears. This exact quantification confirms the theory of **channel substitution**: customers actively self-select into preferred communication mediums while completely ignoring others.

This behavioural polarisation will serve as an excellent mathematical differentiator for the upcoming K-Means clustering algorithm.

Feature Engineering Strategy

Handling Extreme Values in Average Credit Limit

Strategic Context & Purpose

Avg_Credit_Limit spans a substantial numerical range (3,000 to 200,000) coupled with a pronounced rightward skew. Retaining continuous, granular data is essential for accurate modelling, but raw numerical ranges are difficult to translate into actionable business strategy.

To bridge statistical rigour and business application, standard practice dictates a dual-phase approach: **binning** for exploratory analysis and **capping (Winsorization)** for model pre-processing.

Phase 1: Binning for Exploratory Data Analysis

- **Methodology:** converting continuous credit limits into discrete, logical wealth tiers — Low, Medium, High and Premium.
- **Strategic value:** stakeholders need actionable target audiences rather than raw statistical percentiles. Binning turns abstract numerical distributions into a presentation-ready format, letting the Head of Marketing size target segments, assess proportions and forecast the reach of upselling campaigns.

Phase 2: Capping (Winsorization) for Model Optimisation

- **Methodology:** limiting extreme outlier values by capping them at a specific threshold, the 95th percentile, prior to model training.
- **Algorithmic necessity:** distance-based algorithms such as K-Means are highly sensitive to extreme variance because they rely on spatial distance to form segments. Unhandled extremes, such as limits of 200,000, disproportionately influence the algorithm, pulling cluster centroids away and distorting the final segmentation.
- **Strategic value:** capping preserves the numerical format the algorithm requires while neutralising mathematical distortion. Affluent customers remain recognised as the highest tier without their absolute values skewing the Euclidean distance calculations.

Recommended Implementation Workflow

To optimise both stakeholder communication and model performance, a two-step approach is used:

1. **During exploratory data analysis:** engineer a temporary binned feature to construct categorical visualisations, giving leadership a clear, digestible overview of current wealth tier distributions.
2. **During model pre-processing:** discard the categorical binned feature, since distance-based algorithms require strictly numerical inputs. Apply Winsorization to the original Avg_Credit_Limit to manage extreme outliers, followed immediately by standard scaling (Z-score normalisation) to ensure uniform feature weighting during cluster formation.

Customer Wealth Distribution

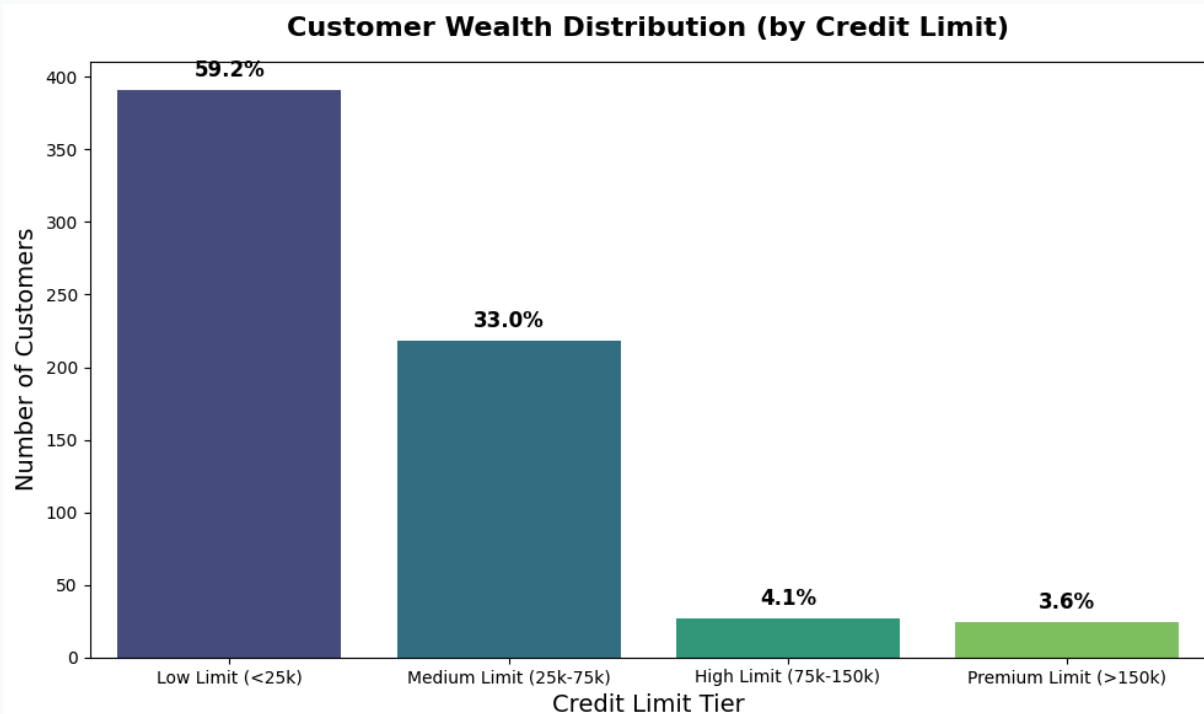


Figure 15: Customer Wealth Distribution by Credit Limit Tier

Key Observations & Strategic Implications

Converting the continuous Avg_Credit_Limit into discrete, business-logical tiers quantifies the portfolio's wealth distribution and provides immediate audience sizing for targeted marketing.

- **The mass-market base (Low Limit):** the clear majority of the portfolio, 59.2% of customers, falls into the lowest credit tier under 25k. This is the baseline, mass-market segment of the bank's credit card users.
- **The prime upsell target (Medium Limit):** exactly 33.0% of customers hold medium limits of 25k to 75k, a large and established middle class. For the Head of Marketing this is the primary, most viable audience for next-tier upselling campaigns.
- **The exclusive minority (High & Premium):** the top two tiers combined represent roughly 8% of the customer base, at 4.1% and 3.6% respectively. Small in headcount, this affluent segment likely commands a disproportionate share of spending power and requires a high-touch retention strategy rather than broad marketing.

Business Takeaway

The portfolio is heavily bottom-weighted, with over 92% of customers holding credit limits under 75,000. Any marketing campaign or service upgrade designed for the "average" customer must be tailored to these lower two tiers, while premium offerings should be explicitly gatekept and marketed exclusively to the top 8%.

Bivariate Analysis

Correlation Heatmap of Numerical Features

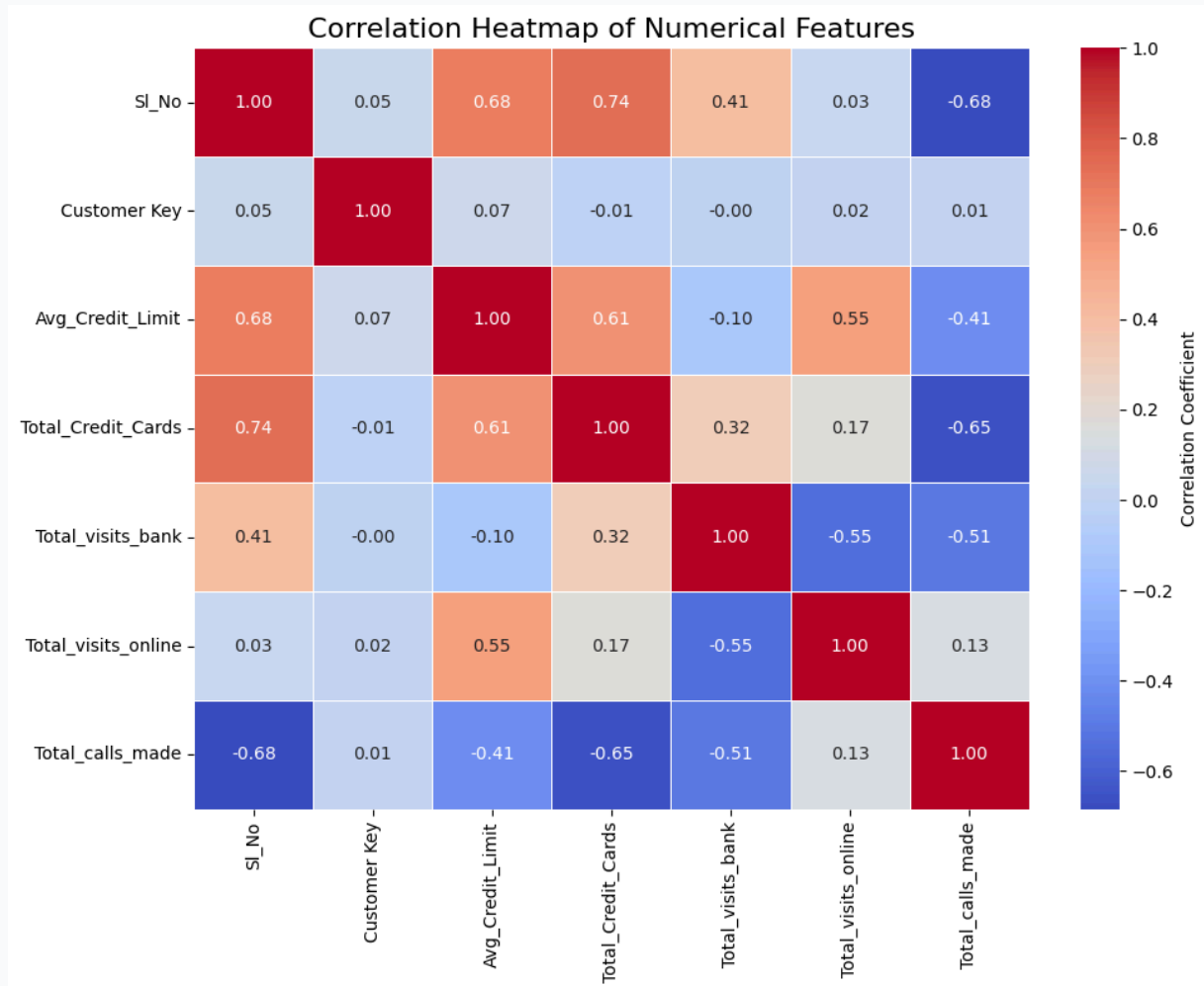


Figure 16: Correlation Heatmap of Numerical Features

Bivariate Boxplots: Wealth Tier vs. Channel Engagement

Strategic Context

Building on the earlier univariate analysis, the custom Credit_Tier feature (Low, Medium, High, Premium) is now used to evaluate how different wealth segments interact with the bank's service channels.

Business Value & Application

This bivariate analysis bridges data science and business strategy by answering a direct, high-value question for the Head of Marketing and the Head of Delivery:

Do our premium customers demand high-touch physical branch services, or do they self-select exclusively into the digital portal?

Plotting each engagement channel against these wealth tiers determines visually whether channel preference scales with customer value. The answer directly informs how service delivery models are tailored and how upcoming campaigns are targeted for each financial segment.

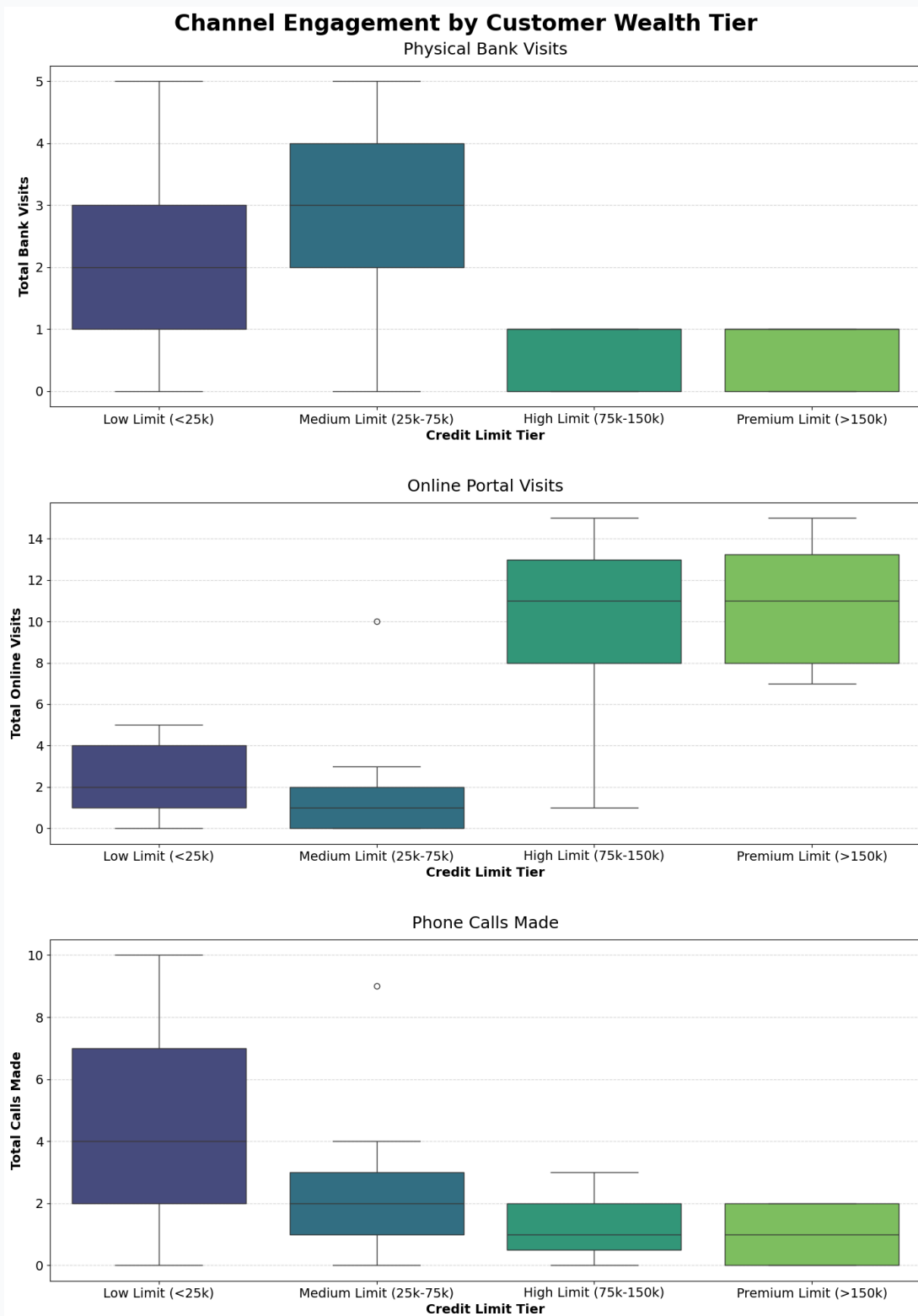


Figure 17: Channel Engagement by Customer Wealth Tier

Key Observations & Strategic Implications

Mapping engagement channels against customer wealth tiers uncovers a massive polarisation in how different financial segments interact with the bank. Channel preference is not random; it scales directly with customer value.

1. Digital Exclusivity Among Premium Customers

- **The affluent shift:** the boxplots reveal a drastic shift in digital adoption as credit limits increase. Customers in the High and Premium tiers show vastly higher online portal visits than the lower tiers.
- **Self-service preference:** this confirms that the bank's most valuable, high-earning customers overwhelmingly prefer the efficiency and autonomy of digital self-service over traditional, high-touch channels.

2. Traditional Channel Dependency for the Mass Market

- **The baseline support needs:** conversely, the Low and Medium credit tiers show significantly lower digital engagement. This mass-market segment remains heavily dependent on traditional channels — physical bank visits and telephonic support — for query resolution and banking needs.

Business Takeaways for Leadership

- **For the Head of Delivery:** the service model must be bifurcated. Investment in premium, white-glove service should be directed entirely toward optimising the digital UI/UX, as this is the primary and often exclusive touchpoint for the highest-value customers. Physical branches and call centres should meanwhile be operationally optimised to efficiently handle the high volume of mass-market enquiries.
- **For the Head of Marketing:** any upselling campaign targeting the High and Premium tiers must be delivered digitally. Campaigns aimed at moving mass-market customers into the Medium tier may still benefit from in-branch promotions or telephonic outreach.

The Density Jointplot: Deep Dive into Channel Substitution

Strategic Context

The correlation heatmap revealed a strong negative correlation (-0.55) between digital engagement and physical branch visits. To demonstrate exactly what this channel substitution looks like in reality for the Head of Delivery, a hexbin jointplot is used.

Why this Visualisation Matters

Unlike a standard scatterplot, which becomes unreadable through overlapping points, a hexbin plot maps customer density mathematically. It combines a spatial distribution with marginal density histograms on the edges for a clearer picture of behaviour.

- **Density mapping:** hexagons darken progressively where a higher concentration of customers sits.
- **Visualising substitution:** the method highlights that the darkest, most heavily populated clusters hug the zero-axes, providing immediate visual proof that high-frequency digital users record exactly zero bank visits.

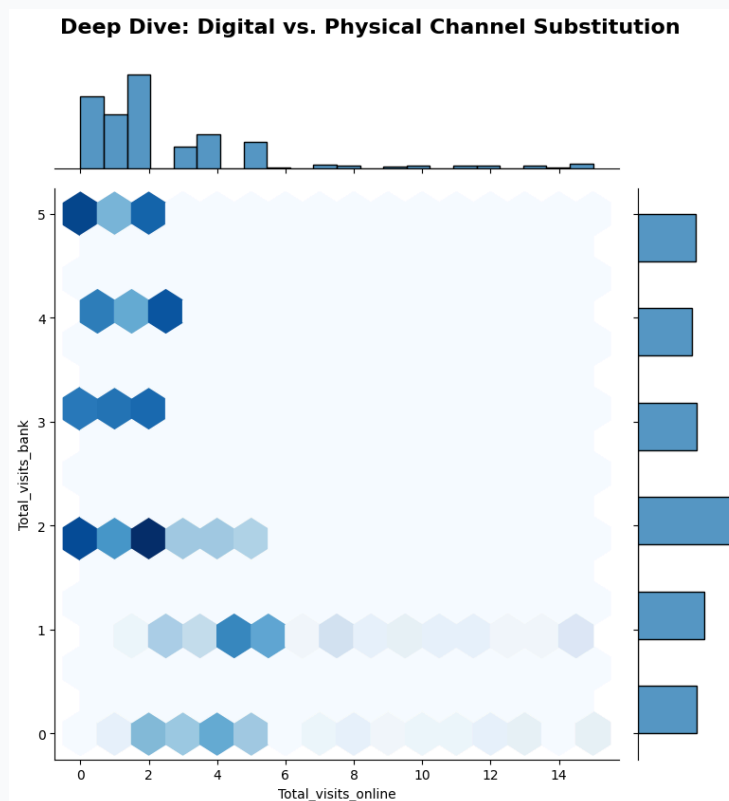


Figure 18: Digital vs. Physical Channel Substitution

Observations and Business Takeaways

- **Visualising the zero-axis anchor:** the darkest, most heavily populated customer clusters are anchored directly against the zero-axes.
- **Absolute substitution:** high-frequency digital users, represented by the extended tail along the x-axis, correspond almost entirely with zero physical branch visits on the y-axis.
- **Strategic implication:** for the Head of Delivery this confirms that the transition to digital banking is not supplementary for these customers — it is a complete substitution. Upgrading digital service delivery will capture and satisfy this highly engaged cohort without requiring concurrent physical branch support.

Outlier Dashboard

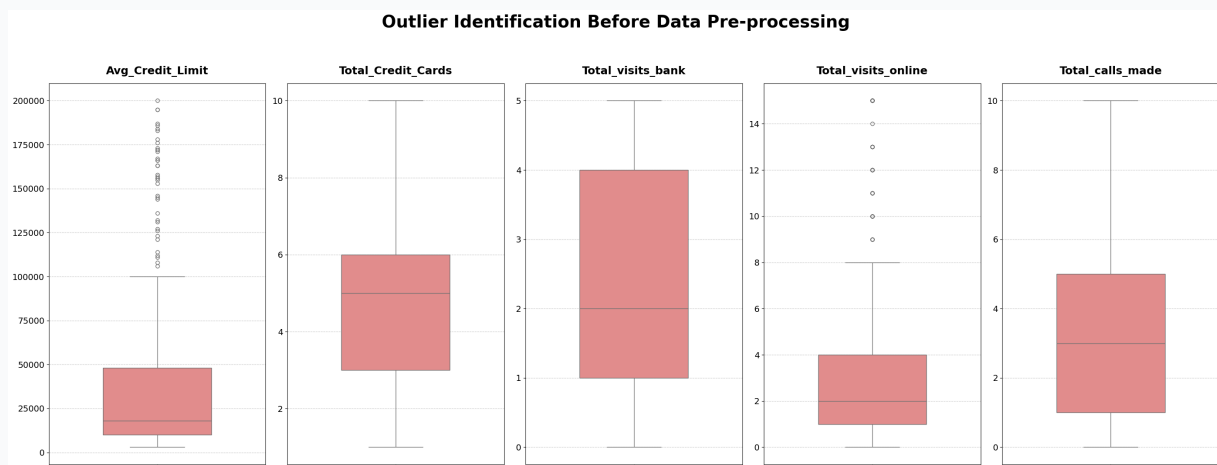


Figure 19: Outlier Identification Before Data Pre-processing

Outlier Identification & Pre-processing Strategy

As a final step in exploratory analysis, the distribution of extreme values across all numerical features must be evaluated. Distance-based algorithms such as K-Means are highly sensitive to statistical outliers and varying magnitudes, which can artificially distort segment formation.

1. Identification of Extreme Values

- **Avg_Credit_Limit:** contains a significant number of high-end outliers extending up to 200,000.
- **Total_visits_online:** shows a prominent long tail of outliers stretching to 15 visits, representing highly engaged digital power users.
- **Stable features:** Total_Credit_Cards, Total_visits_bank and Total_calls_made are tightly bounded, without major statistical outliers pulling away from the upper whiskers.

2. The Magnitude Discrepancy (Scale)

- **Disparity in numerical scale:** the y-axes across these plots highlight a massive disparity. Avg_Credit_Limit operates in the hundreds of thousands, while every other engagement metric operates on a constrained scale of 0 to 15.

Pre-processing Action Plan

1. **Outlier treatment (capping):** apply a capping strategy, bringing extreme values down to the 95th percentile for Avg_Credit_Limit and Total_visits_online. This prevents extreme individuals acting as gravitational "black holes" that pull cluster centroids away from the core customer base.
2. **Standardisation:** scale the entire dataset with a StandardScaler so that a variance of one online visit carries exactly the same mathematical weight as a variance of 10,000 in credit limit.

Use Appropriate Visualisations to Identify Patterns and Insights

The "Cluster Hunter" Pairplot

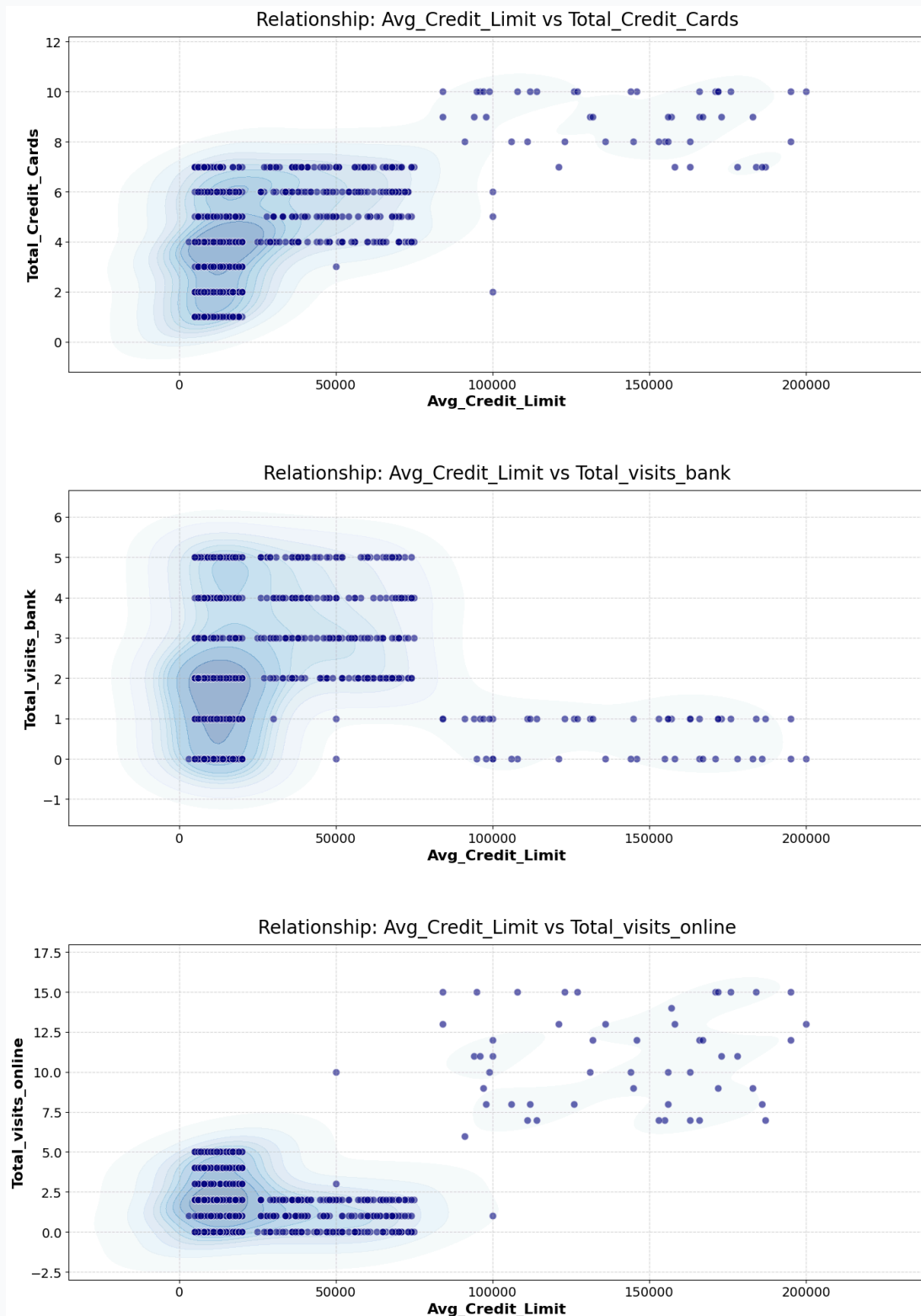


Figure 20: Detailed Bivariate Relationships (1 of 4)

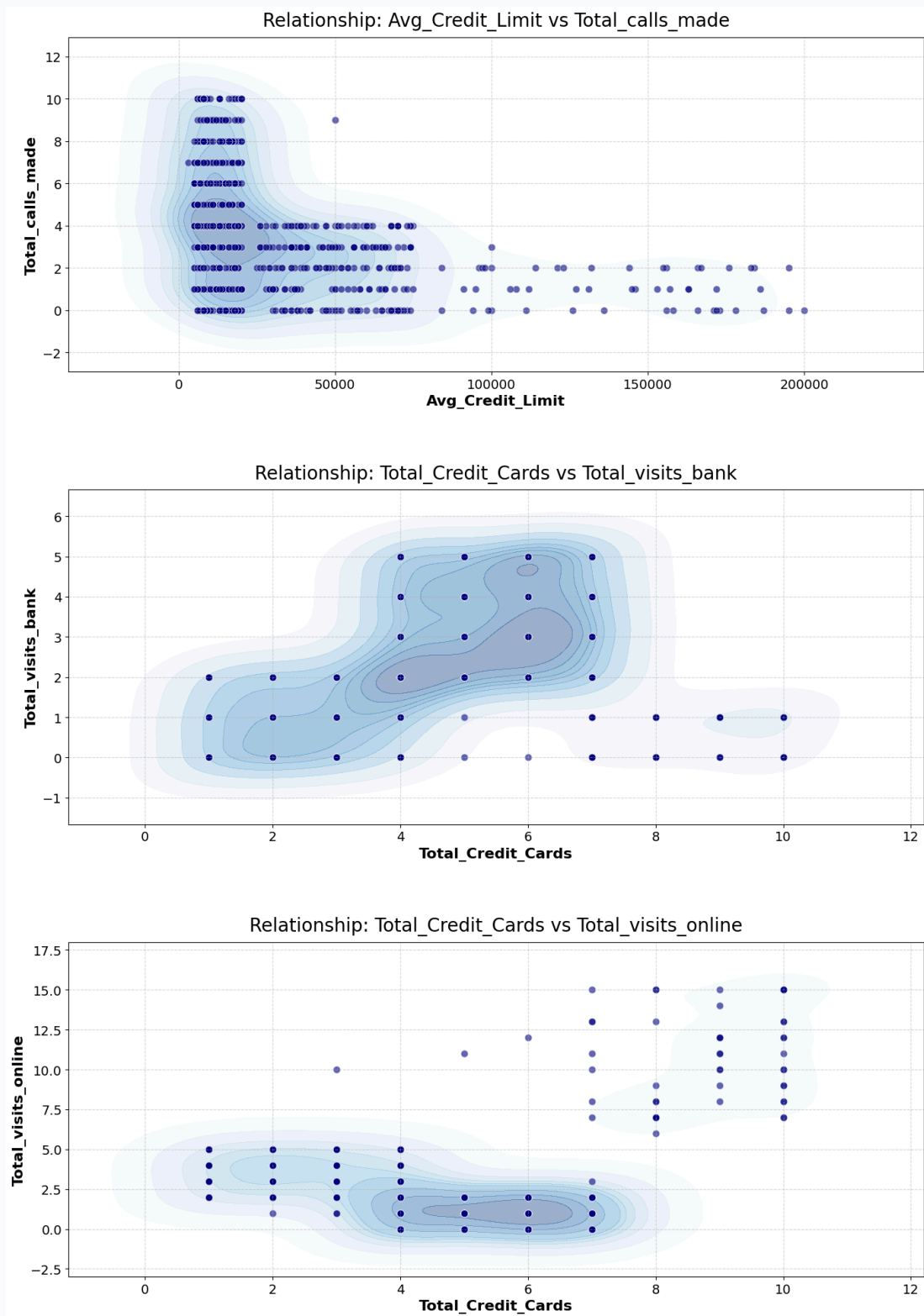


Figure 21: Detailed Bivariate Relationships (2 of 4)

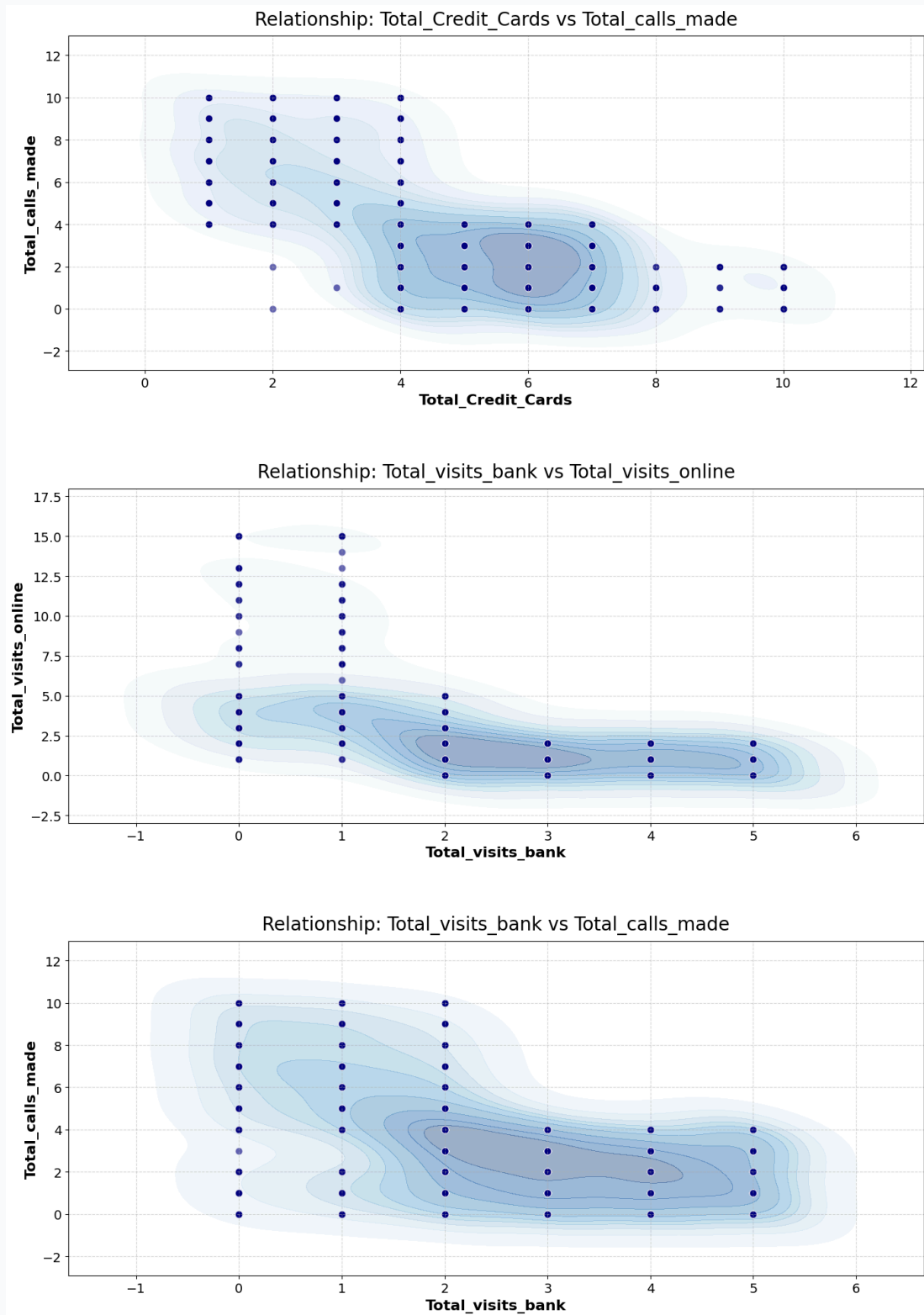


Figure 22: Detailed Bivariate Relationships (3 of 4)

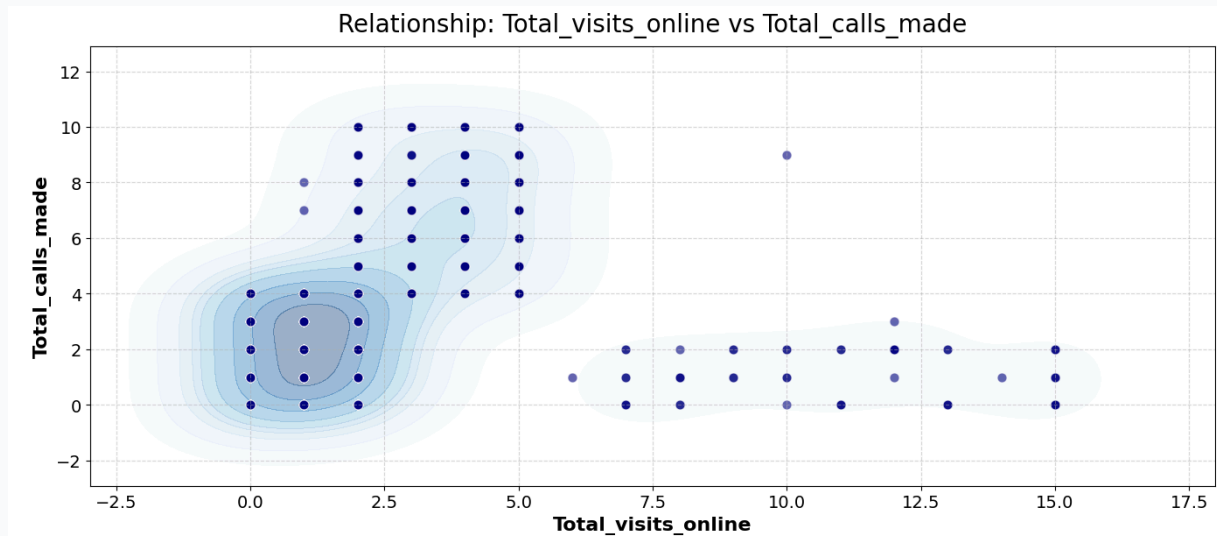


Figure 23: Detailed Bivariate Relationships (4 of 4)

Key Observations: Pairplot Analysis

The stacked pairplot goes beyond correlation values by revealing the shape, density and distribution of customer behaviour. Combining density contours with scatterplots lets us confirm distinct segments visually before applying any algorithm.

- **Visualising natural segments:** the density contours form multiple distinct "puddles" rather than one uniform cloud — strong visual proof that the K-Means and hierarchical algorithms will find natural, separable segments.
- **Confirming product synergy:** Avg_Credit_Limit against Total_Credit_Cards trends upward, but the bulk of the portfolio anchors bottom-left at low limits and few cards. Customers stretching top-right, holding seven to 10 cards at high limits, form a distinct affluent sub-segment structurally separate from the core.
- **Visual proof of channel substitution:** across the channel pairs the points aggressively hug the zero-axes. The cluster reaching 15 online visits sits almost entirely at 0 for bank visits and calls, proving customers select one primary channel and abandon the alternatives.

- **Outlier isolation:** the scatter layer isolates extremes such as the 200,000 credit limits and 15-visit power users, sitting far from the density hotspots. This reinforces the necessity of capping and scaling; unscaled, they act as black holes distorting centroid calculations.
- **The traditional channel overlap:** Total_visits_bank against Total_calls_made shows less extreme polarisation, with a concentrated central blob at one to two branch visits and two to four calls. The non-digital segment uses both traditional channels concurrently rather than picking one exclusively.
- **Product holding vs channel engagement:** the highest-frequency digital users, at 10 to 15 visits, do not sit in the lowest wealth tiers, suggesting that as customers acquire more products or higher limits their preference shifts toward digital self-service.

Synthesis of Key Patterns and Business Insights

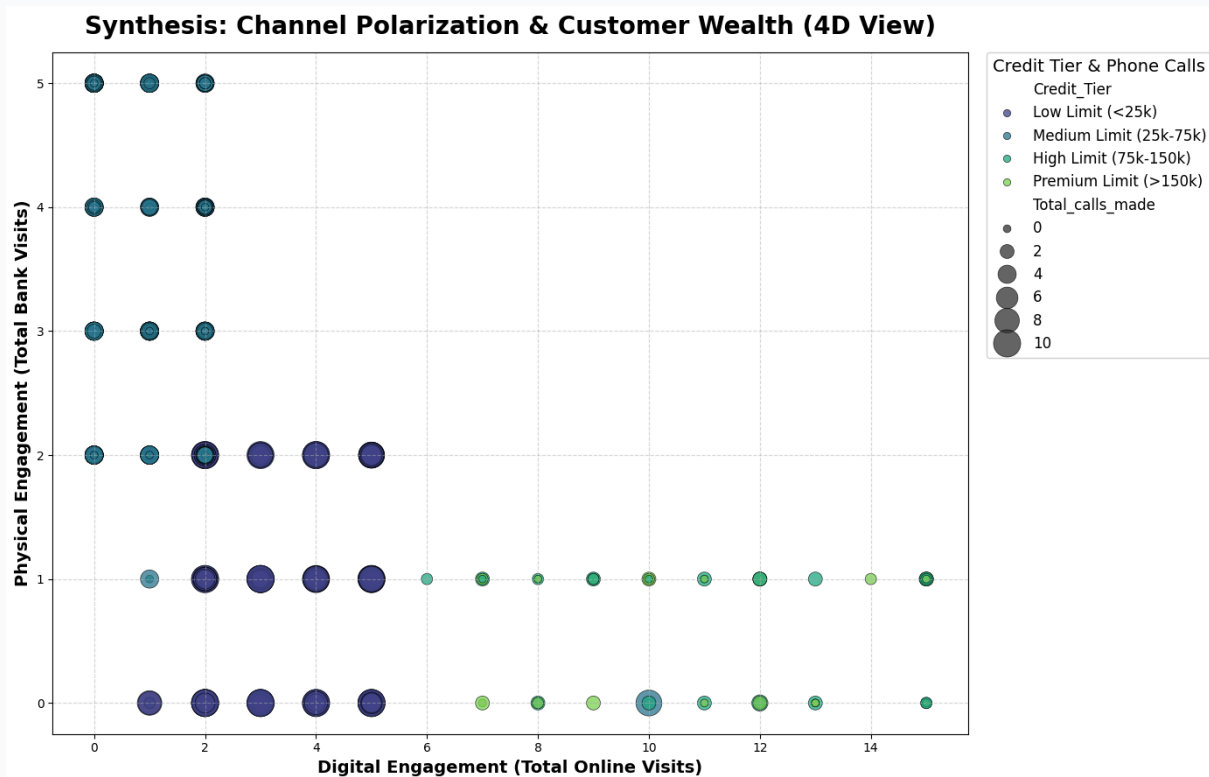


Figure 24: Channel Polarisation & Customer Wealth (4D View)

The multi-dimensional visualisation opposite is the culmination of the exploratory analysis, combining four distinct features — online visits, bank visits, phone calls and wealth tier — to definitively identify the behavioural patterns within the portfolio.

1. The Law of Channel Substitution

- **Strict polarisation:** the data aggressively hugs the X and Y axes. Customers far out on the X-axis record exactly zero physical bank visits, while those clustered along the Y-axis at two to five branch visits entirely bypass the online portal. These channels are highly cannibalistic: customers adopt one primary medium and abandon the others.

2. Wealth-Driven Digital Adoption

- **A stark demographic divide:** the Credit_Tier colouring shows High and Premium customers anchored almost exclusively on the far right of the X-axis, proving the bank's most affluent customers rely entirely on digital self-service. The physical branch cluster along the Y-axis is occupied almost exclusively by the Low and Medium credit tiers.

3. Telephonic Support as a Mass-Market Safety Net

- **Call volume by tier:** bubble size represents Total_calls_made. The largest bubbles concentrate along the Y-axis and near the origin, among the Low and Medium wealth tiers. The High and Premium digital users on the far right show very small bubbles, indicating they resolve queries online without needing verbal support.

Strategic Conclusion for Clustering

The data is deeply fractured into distinct behavioural islands: a high-wealth digital-exclusive cohort, a mass-market in-person cohort, and a high-maintenance telephonic cohort. These natural groupings confirm that K-Means clustering will partition this customer base into distinct, actionable profiles for the Marketing and Operations teams.

Key Meaningful Observations

1. Observations on Individual Variables (Univariate Insights)

- **The zero-engagement phenomenon:** across all communication channels there is a massive cohort recording exactly zero interactions. Customers do not use bank resources evenly; they completely abandon specific channels.
- **Wealth distribution is highly skewed:** Avg_Credit_Limit is heavily right-skewed. The mass market dominates, with roughly 59% of customers holding limits under 25,000, alongside a distinct and highly valuable long tail reaching 200,000.
- **Product holding is concentrated:** Total_Credit_Cards shows a stable, predictable spread. The absolute mode is exactly four cards, with the vast majority holding between three and seven products.

2. Observations on Relationships Between Variables (Bivariate Insights)

- **The law of channel substitution:** a definitive negative correlation exists between digital engagement and in-person banking. Customers are highly polarised: heavy online users record zero branch visits and vice versa, substituting one channel entirely rather than using both.
- **Wealth-driven digital adoption:** channel preference scales directly with customer value. The highest credit tiers overwhelmingly dominate online portal usage, while the mass-market segments depend on branch visits and phone support.
- **Product synergy:** a strong positive correlation links Avg_Credit_Limit and Total_Credit_Cards; the bank is successfully cross-selling multiple credit products to its most creditworthy, high-limit customers.

Strategic Conclusion for Clustering Formulation

The data does not form a single, uniform cloud; it is deeply fractured into distinct behavioural islands, such as high-wealth digital-exclusive versus mass-market in-person. These extreme variances confirm that K-Means clustering will be highly successful in partitioning this customer base.

However, because of the massive scale disparities — credit limits reaching 200,000 while bank visits max out at 5 — outlier capping and standard scaling are strictly mandatory before feeding this data into the clustering algorithm.

7. Data Preprocessing

Missing Value Treatment

Attribute	Missing Values
SI_No	0
Customer Key	0
Avg_Credit_Limit	0
Total_Credit_Cards	0
Total_visits_bank	0
Total_visits_online	0
Total_calls_made	0
Credit_Tier	0

Table 3: Missing Values Audit

- **Observation:** a full audit confirms zero missing or null values across every column in the dataset.
- **Action plan:** because every customer record is completely populated, no imputation technique (mean or median replacement, K-Nearest Neighbours) is required. We proceed directly to outlier treatment and feature scaling.

Outlier Detection and Treatment

As identified in the outlier dashboard, the dataset contains extreme values, most notably in Avg_Credit_Limit (reaching 200,000) and Total_visits_online (reaching 15).

- **Why we cannot ignore them:** K-Means relies on Euclidean distance to form segments. Unhandled extremes act as mathematical black holes, pulling cluster centroids away from the main customer base and distorting the final segments.
- **Why we cannot drop them:** these outliers represent genuine, highly affluent and highly engaged customers. Dropping the rows would mean losing the most valuable premium segment.
- **The solution (Winsorization):** extreme upper values are capped at the 95th percentile. This pulls outliers inward so they no longer distort the distance calculations, while keeping those high-value customers in the dataset.



Figure 25: Distributions After 95th Percentile Capping

Post-Outlier Treatment (Winsorization)

1. Successful mathematical capping

- **Avg_Credit_Limit:** the maximum has been reduced from an extreme 200,000 down to 121,100.
- **Total_visits_online:** the maximum has been reduced from a high of 15 down to roughly 9 visits.

2. Visual distribution improvement

- **Tighter spread:** the post-treatment boxplots show a significantly tighter distribution with the extreme long tails eliminated. These high-value customers are still retained, but their magnitude will no longer artificially distort the Euclidean distance calculations during clustering.

3. Next step: feature scaling (mandatory)

- **Remaining scale disparity:** credit limits are still in the 100,000s while bank visits max out at 5. To ensure all features contribute equally to cluster formation, standard feature scaling must be applied before building the model.

Feature Selection: Removing Non-Predictive Variables

Distance-based algorithms evaluate the mathematical distance between all supplied variables, so including irrelevant or non-numerical data will ruin the cluster formations. Two groups of columns are therefore removed before modelling.

- **Dropping administrative identifiers:** SI_No and Customer Key are system-generated identifiers with no behavioural or financial significance. Left in place, the algorithm would search for patterns in random sequential numbers and distort the segments.
- **Dropping temporary EDA features:** Credit_Tier was engineered as a categorical aid for stakeholder visualisation. Distance algorithms require continuous numerical data, so it is dropped in favour of the original, now outlier-capped, Avg_Credit_Limit.

Final Features Ready for Scaling & Clustering

Original dataset shape: **(660, 8)** → modelling dataset shape: **(660, 5)**

- Avg_Credit_Limit
- Total_Credit_Cards
- Total_visits_bank
- Total_visits_online
- Total_calls_made

Key Observations: Feature Selection

- **Dimensionality reduction:** the dataset was refined from eight features down to the five core attributes necessary for clustering.
- **Removal of mathematical noise:** administrative identifiers were intentionally dropped, preventing random sequential numbers from introducing noise into the distance calculations.
- **Targeted feature set:** the remaining five features encapsulate both the financial capacity and the channel engagement behaviour of each customer. The dataset is now structurally clean and exclusively numerical.

Creating Derived Features (Channel Engagement Ratios)

Raw interaction counts tell the model about the *volume* of a customer's activity. Creating derived ratios shifts the model's focus from volume to behavioural *preference*.

- **Volume vs preference:** Customer A (10 online visits, 2 bank visits) and Customer B (5 online, 1 bank) look far apart on raw counts. Yet both score 83% on a Digital_Engagement_Ratio of online visits divided by total interactions: they share the same digital-first persona.
- **Crystallising channel substitution:** the EDA showed a massive polarisation, with customers substituting one channel for another. A single composite metric feeds that business insight explicitly into the K-Means algorithm.
- **The result:** this engineered feature acts as a powerful mathematical magnet, cleanly pulling digital-heavy customers into one cluster and traditional-heavy customers into another, regardless of how frequently they interact with the bank overall.

Newly Engineered Features Preview

	Total visits bank	Total visits online	Total calls made	Total Interactions	Digital Engagement Ratio
0	1	1.00	0	2.00	0.499998
1	0	9.05	9	18.05	0.501385
2	1	3.00	4	8.00	0.375000
3	1	1.00	4	6.00	0.166666
4	0	9.05	3	12.05	0.751037
5	0	1.00	8	9.00	0.111111
6	0	9.05	2	11.05	0.819004
7	0	1.00	1	2.00	0.499998
8	0	2.00	2	4.00	0.499999
9	0	1.00	7	8.00	0.125000

Table 4: Preview of Engineered Interaction Features

Key Observations: Engineered Derived Features

- **Successful feature creation:** two new composite features were calculated — `Total_Interactions`, an aggregate volume across all touchpoints, and `Digital_Engagement_Ratio`, the share of those interactions occurring online. A small constant ($1e-5$) prevented division-by-zero errors.
- **Quantifying channel substitution:** converting raw counts into a normalised ratio mathematically formalises the substitution behaviour observed during EDA, defining channel preference independently of overall interaction volume.
- **Enhanced clustering potential:** relying on raw counts alone, K-Means might separate high-volume from low-volume digital users despite their shared behaviour. The ratio teaches the model to group by behavioural persona, producing far more actionable segments.

Skewness Mitigation

Distance-based clustering algorithms assume the underlying data is roughly spherical and normally distributed.

- **The issue (shape vs scale):** standard scaling fixes the magnitude of the data but not its shape. `Avg_Credit_Limit` and `Total_visits_online` are heavily right-skewed; left as-is, the long tail of high values pulls cluster centroids off-centre and produces poorly defined segments.
- **The solution:** a log transformation compresses the long tail of high values and spreads out the densely packed lower values, forcing the distribution into a more symmetrical, bell-shaped curve.
- **Technical note:** we use `np.log1p`, which calculates $\log(1 + x)$, rather than a standard log. This is strictly required because features such as `Total_visits_online` contain zeros, and the log of 0 is mathematically undefined.

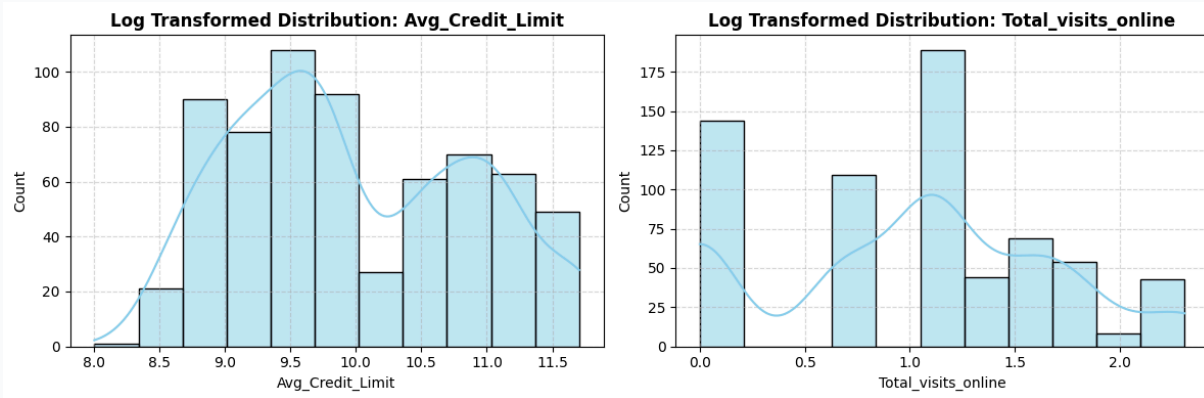


Figure 26: Distributions After Log Transformation

Post-Log Transformation

- **Successful shape normalisation:** for Avg_Credit_Limit the transformation has compressed the severe rightward skew, pulling the trailing tail inward into a far more symmetrical, bell-shaped curve. This spherical shape is mathematically ideal for distance-based algorithms.
- **Variance compression:** Total_visits_online still exhibits distinct peaks due to the discrete nature of the counts, particularly the large zero-visit cohort, but the extreme spread at the high end is significantly reduced.
- **Centroid stabilisation:** with the underlying shape normalised, the long tails no longer act as gravitational weights pulling centroids off-centre. With both outliers and skewness addressed, the dataset is primed for standard scaling.

Data Scaling (Standardisation)

Distance-based clustering algorithms rely on calculating the Euclidean distance between data points, which makes them highly sensitive to the magnitude of the features.

- **The problem of magnitude:** Avg_Credit_Limit operates on a massive scale, from 3,000 up to 121,100 post-capping, while the channel engagement features operate on a micro-scale of 0 to roughly 10.
- **The consequence of unscaled data:** fed in as-is, a difference of 5,000 in credit limit would completely overpower a difference of 5 bank visits. The algorithm would effectively ignore the engagement channels and cluster customers on credit limit alone.
- **The solution:** StandardScaler (Z-score normalisation) rescales every feature to a mean of 0 and a standard deviation of 1, ensuring each carries the exact same mathematical weight when forming segments.

Scaled Data Overview

	Avg Credit Limit	Total Credit Cards	Total visits bank	Total visits online	Total calls made	Total Interactions	Digital Engagement Ratio
0	1.698631	-1.249225	-0.860451	-0.468673	-1.251537	-2.028140	1.110907
1	0.929520	-0.787585	-1.473731	1.914295	1.891859	3.046462	1.117250
2	0.929520	1.058973	-0.860451	0.554445	0.145528	-0.131093	0.539424
3	0.362718	0.135694	-0.860451	-0.468673	0.145528	-0.763442	-0.413063
4	1.698631	0.597334	-1.473731	1.914295	-0.203739	1.149414	2.258644

Table 5: Scaled Data Overview (mean $\approx$ 0, standard deviation $\approx$ 1)

Key Observations: Post-Scaling Transformation

- **Successful standardisation:** the StandardScaler has transformed every numerical feature into a Z-score. Regardless of original unit or magnitude, all features now share a mean of approximately 0 and a standard deviation of 1.
- **Interpreting the Z-scores:** each value represents how many standard deviations a data point sits from that feature's average. In row 0, an Avg_Credit_Limit of 1.70 places that customer well above the portfolio average, while a Total_calls_made of -1.25 shows below-average phone engagement.
- **Model readiness:** the scale disparity between financial metrics (previously reaching 121,100) and engagement metrics (capping at 5 to 15) has been entirely eliminated. Euclidean distance calculations will now group customers on holistic behavioural patterns rather than being dominated by high-magnitude variables.

8. Model Building: K-Means Clustering

Overview

K-Means is an unsupervised algorithm that partitions a dataset into K distinct, non-overlapping clusters by grouping records on multi-dimensional feature similarity. It works on unlabelled data, and its objective is twofold: minimise the variance inside each cluster while maximising the distance between clusters.

Methodology: Finding the Optimal K

K-Means cannot infer how many natural segments exist in the data. K is a hyperparameter that has to be fixed before training, so rather than choose it arbitrarily we evaluated models across $K = 1$ to 9 against two complementary metrics.

1. The Elbow Method (Inertia / Within-Cluster Sum of Squares)

- **What it measures:** the total variance within each cluster, i.e. how tightly points sit around their centroid. Lower inertia means denser clusters.
- **What to look for:** inertia always falls as K rises (with one cluster per customer it would reach zero), so the signal is the "elbow" in the curve — the point of diminishing returns, beyond which extra clusters stop meaningfully improving compactness.

2. Silhouette Analysis

- **What it measures:** cohesion (closeness to one's own cluster) weighed against separation (distance from the nearest neighbouring cluster).
- **What to look for:** scores run from -1 to $+1$. Near $+1$ the point is cleanly assigned and well separated; near 0 the clusters overlap. The best K is the highest peak in the average score.

The Synthesis Approach

No single metric decides K. A robust model sits at the elbow of the inertia curve, holds a high silhouette score at the same time, and produces a number of segments the business can actually act on.

Plotting the Elbow Curve

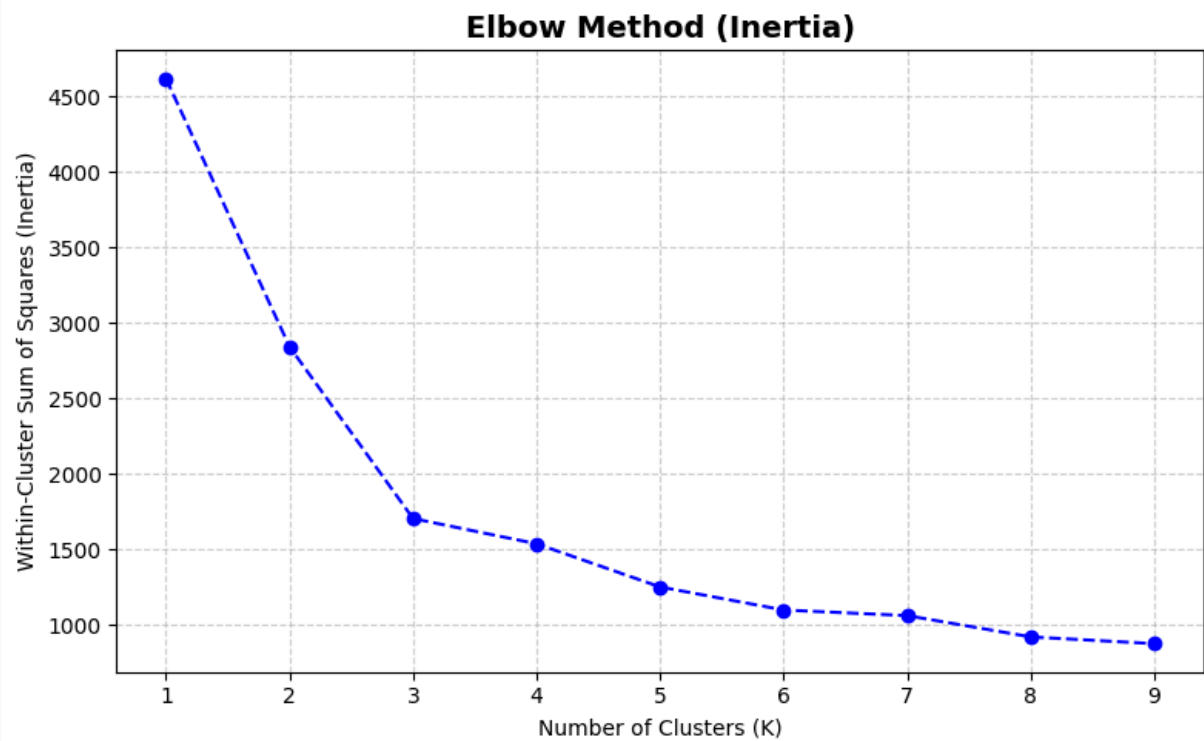


Figure 27: Elbow Method — Within-Cluster Sum of Squares by K

Key Observations

- **Inertia trajectory:** total inertia falls as K increases across the 1–9 range, reflecting progressively tighter groupings.
- **The inflection point:** the decline is steep from K = 1 to K = 3, and at exactly K = 3 the curve forms a distinct elbow.
- **Diminishing returns:** beyond K = 3 the curve flattens. The further reduction at K = 4 or K = 5 is marginal, so additional clusters buy over-fragmentation rather than compactness.
- **Conclusion:** on the elbow evidence alone, K = 3 is the soundest choice — the best balance between low intra-cluster variance and low complexity.

Checking Silhouette Scores

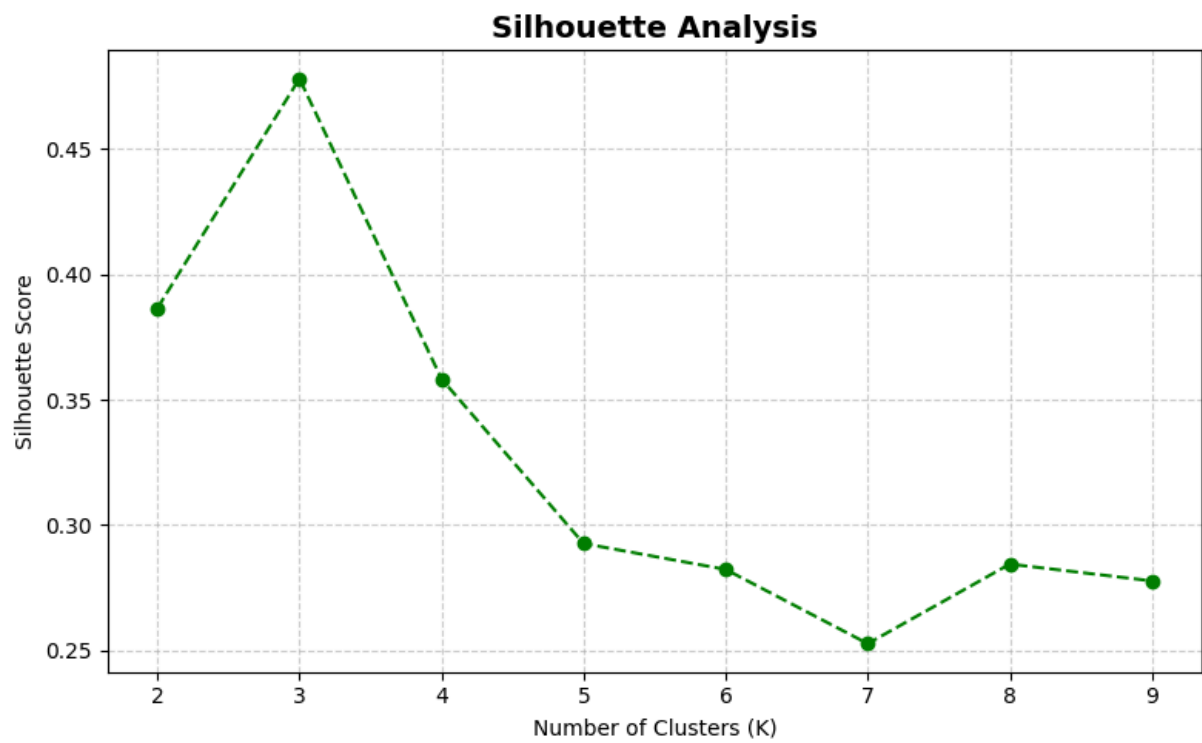


Figure 28: Silhouette Analysis — Average Score by K

Key Observations

- **Where the peaks are:** the strongest scores sit at the low end of the range, at $K = 2$ and $K = 3$.
- **The drop-off:** beyond $K = 3$, and sharply after $K = 4$, the score falls below 0.30 — confirming that five or more segments produce heavy overlap and poorly defined boundaries.
- **Conclusion:** $K = 2$ peaks marginally higher on this dataset, but $K = 3$ holds a robust score just before the collapse and gives a far richer strategic breakdown than a binary split. Read alongside the elbow curve, $K = 3$ is validated.

Silhouette Visualizer: Reading the Plot

Average scores tell us whether a model is broadly sound; the Silhouette Visualizer acts as an X-ray on each individual segment.

- **The x-axis (fit score):** how well each customer fits its segment. 1.0 is a perfect fit, 0 is borderline, and a negative value means the customer was probably placed in the wrong group.
- **The coloured shapes (segments):** each shape is one cluster; its vertical thickness is the customer count. Balanced thickness is the goal, not one dominant group beside a nearly empty one.
- **The red dashed line (benchmark):** the average fit score for the model. A segment counts as strong only if its shape extends past this line.

Why Negative Scores Matter

A negative score means the point sits closer to a neighbouring centroid than to its own. In business terms these are "confused" customers on the blurry boundary between two segments, whose banking behaviour is not distinct enough to place them confidently. Over-splitting the data ($K = 4$ and above) sharply increases these misclassifications, so the winning model is the one with evenly sized segments that all extend past the average line and carry few or no negative scores.

Silhouette Visualizer: $K = 3$ versus $K = 4$

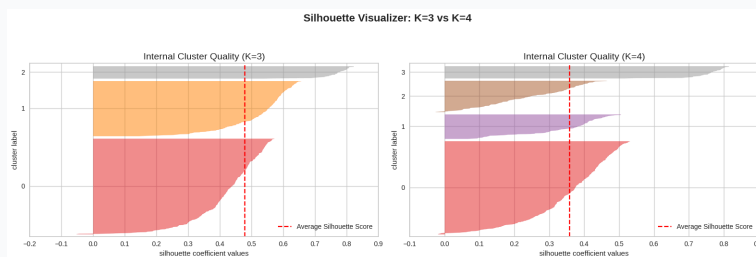


Figure 29: Internal Cluster Quality — $K = 3$ compared with $K = 4$

Key Observations

- **Cluster balance:** the three $K = 3$ shapes have fairly uniform thickness, so customers are spread evenly. At $K = 4$ the segments fracture into smaller, less meaningful micro-segments.
- **Surpassing the average:** all three $K = 3$ clusters extend comfortably past the red average line, confirming each is mathematically strong.
- **Misclassifications:** $K = 3$ holds clean boundaries with almost no negative overlap, whereas adding a fourth cluster introduces instability.
- **Conclusion:** $K = 3$ produces three robust, evenly distributed and genuinely distinct segments.

Variance Decomposition: TSS, WSS and BSS

Total Sum of Squares (TSS) is fixed at 4,620 for this dataset. Each value of K splits it between variance still trapped inside the clusters (WSS) and variance successfully explained by the separation between them (BSS).

Clusters (K)	WSS (Inertia)	BSS (Between SS)	WSS / BSS Ratio	Total SS	% Variance Explained
1	4620.00	0.00	N/A	4620.0	0.00%
2	2842.64	1777.36	1.599	4620.0	38.47%
3	1703.82	2916.18	0.584	4620.0	63.12%
4	1536.28	3083.72	0.498	4620.0	66.75%
5	1250.24	3369.76	0.371	4620.0	72.94%
6	1097.44	3522.56	0.312	4620.0	76.25%
7	1060.32	3559.68	0.298	4620.0	77.05%
8	919.05	3700.95	0.248	4620.0	80.11%
9	874.34	3745.66	0.233	4620.0	81.07%

Table 6: Variance Decomposition and WSS / BSS Ratio by Number of Clusters

Key Observations

- **Variance captured at K = 3:** splitting the portfolio into three segments explains 63.12% of total variance in customer behaviour.
- **The trade-off:** more clusters always capture more variance (81.07% at K = 9), but the gain comes as hyper-fragmented micro-segments that cannot be marketed to at scale.
- **The WSS / BSS ratio:** dividing within-cluster by between-cluster variance gives a single measure of cluster quality. It collapses from 1.599 at K = 2 to 0.584 at K = 3, then only creeps downward — tight cohesion and strong separation are already achieved at three clusters.

Final Conclusion on the Optimal K

Three independent checks converge on the same answer. The **Silhouette Visualizer** shows dense, evenly balanced segments with virtually no misclassification; **variance analysis** confirms a majority of total variance is captured; and the **WSS / BSS ratio** stabilises at K = 3. K = 3 is therefore selected as the final hyperparameter, yielding three distinct and actionable customer personas.

Final Model Execution

This step moves the project from mathematical validation into business profiling.

- **Algorithmic execution:** the final K-Means model was initialised with the validated hyperparameter $K = 3$ and trained on the strictly numerical, standardised dataset (`df_scaled`).
- **Feature exclusion:** categorical features engineered purely for EDA visualisation, such as `Credit_Tier`, were withheld from training. K-Means depends on continuous Euclidean distance, so the financial grouping was derived from the continuous `Avg_Credit_Limit` instead.
- **Customer assignment:** all 660 customers were partitioned into three behavioural segments, labelled 0, 1 and 2.
- **Data integration:** the labels were mapped back onto the original, unscaled dataset so profiling reports real currency amounts and true interaction counts rather than z-scores.

SI_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Credit_Tier	Cluster
548	38125	26000.0	4	5	2.0	4	Medium Limit (25k–75k)	0
354	94437	9000.0	5	4	1.0	3	Low Limit (<25k)	0
500	65825	68000.0	6	4	2.0	2	Medium Limit (25k–75k)	0
174	38410	9000.0	2	1	5.0	8	Low Limit (<25k)	1
242	81878	10000.0	4	5	1.0	3	Low Limit (<25k)	0
342	70779	18000.0	4	3	2.0	0	Low Limit (<25k)	0
648	79953	121100.0	9	0	9.0	2	Premium Limit (>150k)	2
219	28208	19000.0	3	1	5.0	7	Low Limit (<25k)	1
121	16577	10000.0	4	2	4.0	6	Low Limit (<25k)	1
135	31256	13000.0	4	1	5.0	7	Low Limit (<25k)	1

Table 7: Cluster Assignment Preview (labels mapped back to the unscaled data)

Cluster Profiling: Persona Identification

Averaging the original, unscaled feature values within each cluster turns the mathematical partition into three recognisable customer personas.

Cluster	Persona	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Customer_Count
0	Traditional Mass-Market	33,733.85	5.51	3.48	0.98	2.00	387
1	Support-Reliant Base	12,161.43	2.41	0.94	3.56	6.90	223
2	Premium Digital High-Rollers	114,126.00	8.74	0.60	8.57	1.08	50

Table 8: Customer Profiles — Averages by Cluster

Cluster 0 — "The Traditional Mass-Market" (387 customers)

Mid-level limits averaging \$33,734 across 5.5 cards. The most branch-oriented group in the portfolio, at 3.48 in-person visits a year against barely one online login. This is the bank's stable core: moderate spending power, served face to face by preference, and the largest segment at 59% of the book.

Cluster 1 — "The Support-Reliant Base" (223 customers)

The lowest limits, averaging \$12,161, and the fewest products at 2.41 cards. Engagement runs through the call centre at 6.90 calls a year, with moderate online use (3.56) and almost no branch contact (0.94). Operationally heavy: the lowest financial value paired with the highest demand on support resources.

Cluster 2 — "The Premium Digital High-Rollers" (50 customers)

The elite tier, holding limits averaging \$114,126 and 8.74 products — roughly nine times the credit capacity of Cluster 1. Engagement is almost entirely self-service, at 8.57 online visits a year against 0.60 branch visits and 1.08 calls. Only 7.6% of customers, yet by far the most valuable and the least costly to serve.

Visualising the Cluster Profiles



Figure 30: Cluster Comparison — Financial and Channel Engagement Profiles

Key Strategic Observations

The decisive evidence here is that the interquartile ranges — the boxes themselves — barely overlap across the critical metrics. The three personas describe structurally separate groups of customers, not arbitrary cuts through one population.

- **Financial polarisation:** Cluster 2's credit-limit box sits in a band of its own, well clear of the other two. Clusters 0 and 1 together form the mass market, with Cluster 1 compressed at the bottom of the range.
- **Absolute channel substitution:** the bank-visit and online-visit plots confirm the substitution pattern identified during pre-processing. Cluster 2 is digital-only, with near-zero branch visits; Cluster 0 mirrors it exactly in reverse, favouring the branch while abandoning the portal.
- **Operational support division:** the calls plot isolates Cluster 1 as the driver of call-centre load, with a median of seven calls against roughly two for Cluster 0 and one for Cluster 2.
- **Conclusion:** this visual proof completes the validation. Three robust, mathematically distinct personas can now be handed to the Head of Marketing and the Head of Delivery for execution.

Why Total_Credit_Cards Is Not Plotted

Card count is a defining attribute of these clusters, but it was deliberately left out of this grid. The earlier correlation analysis showed it moves closely with Avg_Credit_Limit, so credit limit already serves as the single strongest proxy for financial tier and plotting both would only add redundancy. Restricting the view to four metrics also keeps the grid symmetrical and holds attention on the core narrative: financial capacity combined with channel substitution.

9. Model Building: Hierarchical Clustering

Overview

Hierarchical clustering is a second, highly visual unsupervised approach. Instead of requiring the number of clusters up front, it builds a complete bottom-up hierarchy of the dataset and renders it as a dendrogram, from which the analyst reads off the natural number of groups. Running it alongside K-Means gives an independent check on the segmentation.

Cophenetic Correlation Analysis

Before building the tree we must choose how to measure distance between customers and how to merge groups. The cophenetic correlation compares the distances implied by the dendrogram against the true pairwise distances in the data — the higher the score, the more faithfully the tree represents the underlying structure. Every distance metric was tested against every applicable linkage method on `df_scaled`.

Linkage	Euclidean	Chebyshev	Mahalanobis	Cityblock
Single	0.6487	0.5306	0.6815	0.5966
Complete	0.7939	0.6915	0.6021	0.7681
Average	0.8485	0.8151	0.7564	0.8371
Weighted	0.7967	0.8106	0.7160	0.7866
Centroid	0.8421	—	—	—
Ward	0.7689	—	—	—

Table 9: Cophenetic Correlation by Distance Metric and Linkage Method (centroid and Ward are defined only for Euclidean distance)

Key Observations

- **Winning combination:** Euclidean distance with average linkage scores 0.8485, the highest of the eighteen combinations tested, and is carried forward as the final configuration.
- **Linkage matters more than metric:** average linkage leads its column under all four metrics, while single linkage is the weakest throughout — it chains points into long straggling groups rather than compact ones.
- **Close runner-up:** centroid linkage on Euclidean distance reaches 0.8421, essentially tied with the winner; Ward, despite being the common default, trails at 0.7689.

Dendrograms by Linkage Method

Plotting the tree under each linkage method shows why the cophenetic scores differ so sharply.

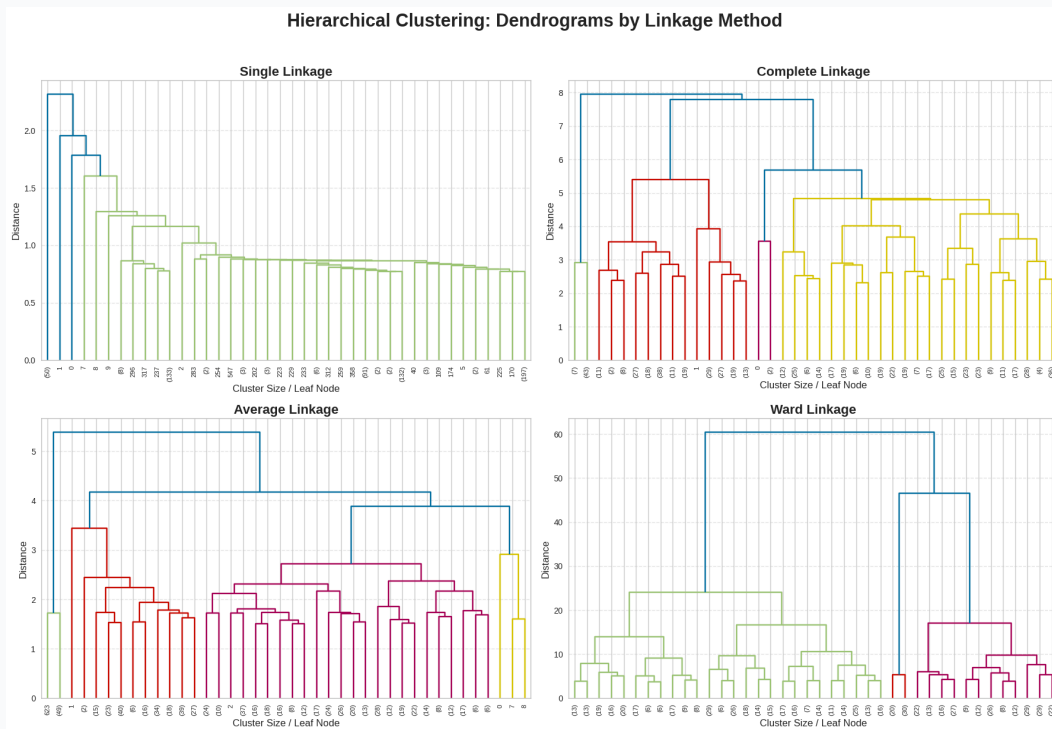


Figure 31: Hierarchical Clustering — Dendrograms by Linkage Method

- **Single linkage** collapses into a long chain of near-identical merge heights, offering no obvious cut point — consistent with its 0.6487 score.
- **Complete and Ward linkage** produce more balanced trees, but their branches merge across a compressed band of distances, making the natural number of groups ambiguous.
- **Average linkage** yields well-separated branches with a clear vertical gap between the last few merges, which is exactly the structure a clean cut requires.

Determining the Optimal Number of Clusters

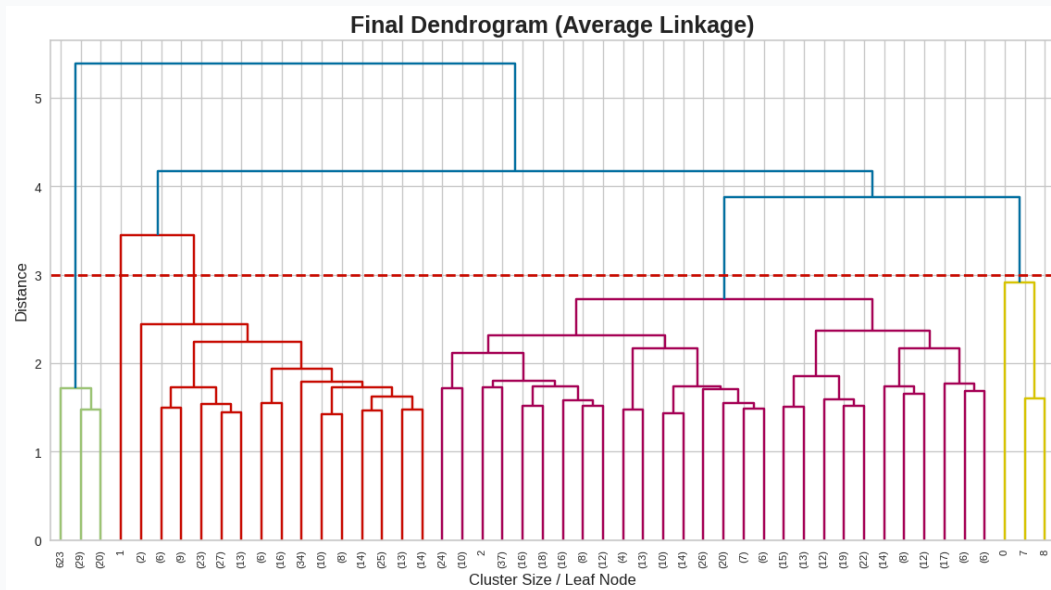


Figure 32: Final Dendrogram (Average Linkage) with Cut Line at Distance = 3.0

Key Observations

- **Reading the tree:** the natural number of segments is found by locating the tallest vertical lines that no horizontal merge crosses. In this dendrogram the longest uninterrupted spans open up just above a distance of 2.0.
- **The threshold:** cutting horizontally through that gap at a distance of 3.0 intersects exactly three vertical branches, so the hierarchy resolves naturally into three segments.
- **Cross-algorithmic agreement:** this matches the K-Means result, where the elbow method and silhouette analysis both landed on $K = 3$. Two algorithms with entirely different mechanics — centroid-based partitioning versus bottom-up agglomeration — arriving at the same answer is strong evidence that the three segments are real structure rather than an artefact of either method.

Hierarchical Cluster Profiling

Profiling the three agglomerative segments on the original, unscaled features reproduces the same three personas found by K-Means.

HC_Cluster	Persona	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
0	Traditional Mass-Market	33,591.26	5.50	3.47	0.99	2.00
1	Premium Digital High-Rollers	114,126.00	8.74	0.60	8.57	1.08
2	Support-Reliant Base	12,217.19	2.40	0.94	3.57	6.93

Table 10: Hierarchical Clustering — Segment Profiles (Averages)

Persona Identification

- **HC Cluster 0 — Traditional Mass-Market:** mid-tier limits averaging \$33,591 across 5.50 cards, with the highest branch footfall at 3.47 visits a year and little digital or telephone activity. Standard mass-market customers who bank in person.
- **HC Cluster 1 — Premium Digital High-Rollers:** the highest limits at \$114,126 and the most products at 8.74 cards, engaging almost entirely through the portal at 8.57 visits a year against near-zero branch visits and calls. Affluent and self-sufficient.
- **HC Cluster 2 — Support-Reliant Base:** the lowest limits at \$12,217 and fewest cards at 2.40, with the heaviest call volume at 6.93 a year and minimal in-person contact. An entry-level segment dependent on telephone support.

Note that the numeric labels differ from the K-Means run — the premium segment is cluster 1 here but cluster 2 there. Label order carries no meaning: K-Means initialises randomly while hierarchical clustering merges bottom-up, so the IDs are assigned in different orders even when the underlying groups are identical.

10. K-Means versus Hierarchical Clustering

Convergence on Three Segments

Both algorithms independently identified exactly three optimal segments, each by its own logic. K-Means was validated numerically, through the inflection point of the inertia curve and the peak of the silhouette score. Hierarchical clustering was validated structurally, through a dendrogram that splits into three uninterrupted branches under a cut at distance 3.0. These are different mathematical routes to the same answer.

Persona Stability Across Algorithms

The stronger test is whether the segments themselves match, not just their count. Setting the two profile tables side by side shows they do.

Persona	K-Means ID	HC ID	Avg limit (K-Means)	Avg limit (Hierarchical)	Difference
Traditional Mass-Market	0	0	33,733.85	33,591.26	-142.59
Premium Digital High-Rollers	2	1	114,126.00	114,126.00	0.00
Support-Reliant Base	1	2	12,161.43	12,217.19	+55.76

Table 11: Persona Comparison — K-Means versus Hierarchical Clustering

- **Identical premium segment:** the high-roller profile is reproduced exactly — \$114,126.00, 8.74 cards, 0.60 branch visits, 8.57 online visits, 1.08 calls. Both algorithms isolated the same 50 customers.
- **Negligible drift elsewhere:** the mass-market average moves by \$143 and the support-reliant average by \$56, differences of well under half a percent on a portfolio whose limits span \$3,000 to \$200,000. A handful of boundary customers changed hands; the segments did not.
- **Channel behaviour holds:** branch, online and call averages agree to within 0.03 across all three personas.

When two algorithms built on different principles converge on the same solution, that convergence is itself evidence. It indicates the three segments reflect genuine customer behaviour rather than an artefact of either method, and the Head of Marketing and Head of Delivery can act on them with confidence.

Computation Time and Scalability

- **K-Means is lightweight:** cost grows roughly linearly with the number of records, because each pass only measures distances from every point to three centroids. It stays fast on very large datasets.
- **Hierarchical clustering is expensive:** it must compute the distance between every pair of points to build the linkage matrix, so both runtime and memory grow quadratically. Cost rises steeply as the dataset grows.
- **On this dataset the difference is invisible:** at 660 observations both algorithms complete essentially instantly, which is precisely why it was practical to run both as a cross-check here.

Deployment recommendation: if AllLife Bank scales this model to hundreds of thousands or millions of live cardholders, K-Means is the only viable production engine. Hierarchical clustering should stay in exploratory research on samples, where the dendrogram earns its keep by validating the number of clusters visually — exactly the role it played in this analysis.

11. Actionable Business Recommendations

The segmentation, confirmed by two independent algorithms, exposes one dominant pattern: extreme channel polarisation. Each segment commits almost entirely to a single method of contact and abandons the rest. Every recommendation below follows from that — matching the offer to the segment, and the delivery to the channel it already uses.

For the Head of Marketing

- **Traditional Mass-Market — upsell in the branch:** the largest segment at 387 customers, on moderate limits (~\$33.7k) and an established 5.5 products each, making it the prime pool for next-tier upgrades. Because they transact almost exclusively in person, arm tellers and branch managers with pre-approved limit-upgrade offers; digital campaigns will not reach them.
- **Premium Digital High-Rollers — retain through the app:** fifty customers carrying limits above \$114k who neither call nor visit, so phone and direct mail are wasted spend and an irritation. Reach them inside the portal and mobile app with premium lifestyle perks, travel rewards and elite status benefits. The objective is retention of the highest-value accounts, not acquisition.
- **Support-Reliant Base — nurture, don't push:** the lowest limits (~\$12.2k) and fewest cards mean aggressive upselling carries elevated credit risk. Lead with foundational products and financial education — balance-transfer promotions, introductory rewards, credit-building tools — offered through the call centre they already use daily.

For the Head of Delivery

- **Digital self-service for the high-rollers:** the bank's most valuable customers use the online portal exclusively, so the digital experience carries the whole relationship. Prioritise automated dispute resolution, instant limit increases and digital card issuance, so this segment never has to fall back on a support line it clearly does not want.
- **Branch modernisation for the mass market:** Cluster 0 generates nearly all physical branch traffic, the costliest channel to serve. Optimise branches for fast query resolution and add digital onboarding kiosks; staff should walk customers through the app for the query that brought them in, migrating routine transactions to self-service over time.
- **Automation for the support-reliant base:** this segment sets call-centre volume, and most of it is routine — balance checks, payment due dates, basic account queries. A smart IVR or AI voice assistant handling those instantly would cut hold times, raise satisfaction and lower operating cost, while freeing agents for the cases that genuinely need a person.

Together these measures address both halves of the original brief: sharper marketing through segment-specific targeting, and a service model that meets each group on the channel it already prefers — directly answering the poor support perception that prompted this analysis.